

Course: BIM FUNDAMENTAL AND APPLICATIONS



POLITECNICO

MILANO 1863

Author:

Haseeb Ali

Student ID:

222834

Submitted to: Prof. Carlo lapige De Gaetani

February 4th, 2024

Contents

1. Introduction:	3
2. Conceptual Design:	4
2.1. Grid and Levels.....	4
2.2. Object Placement.....	5
2.3. Geometrical/Logical Constraints.....	6
2.4. Parametric object properties	7
2.5. Roof by Footprint	8
2.6. Floor by Footprint	9
2.7. New Type Creation.....	9
2.8. Creation of Stairs.....	10
2.9. Topography	11
3. Structural Design:.....	13
3.1. Linking Revit Conceptual model.....	13
3.2. Copy/Monitor elements.....	13
3.3. Changes Detection	15
3.4. Structural Elements.....	16
3.4.1. Structural Columns:.....	16
3.4.2. Structural Beams	16
3.4.3. Beam Systems	17
3.4.4. Structural Foundation	18
3.4.5. Structural Slabs	19
3.5. Objects Layering.....	20
3.6. Work plane definition	20
3.7. Model Line Exploitation	21
3.8. Model in place objects	22
3.9. Reinforcement Design.....	23
3.9.1. Column Reinforcement:	23
3.9.2. Slab Reinforcement (Area).....	25
3.9.3. Beam Reinforcement	26
3.9.4. Shear Wall Reinforcement	27
3.9.5. Foundation Reinforcement:	29
3.10. Rebar Cover Setup	30

3.11.	Analytical model.....	31
4.	Architectural Design:.....	32
4.1.	Loading Families.....	32
4.2.	Material Management	33
4.3.	Object Refinements	33
4.3.1.	Floors:.....	34
4.3.2.	Walls:.....	35
4.4.	Creating Families.....	37
4.5.	Parametrizing Families.....	38
4.6.	Topographic Refinement:	40
5.	Final Design:	43
5.1.	Federated Model Assembly	43
5.2.	Phases Management.....	43
5.3.	Interference Check.....	45
5.4.	Elements Schedule	48
5.5.	Material Takeoff schedule and Costs Computation.....	49
5.5.1.	Concrete:.....	50
5.5.2.	Masonry Units (Concrete):.....	52
5.6.	Rooms Legend and Schedule	53
5.7.	Rendering.....	54
5.8.	Sheets Generation	56

1. Introduction:

The Project comprised of detailing and following the procedures learnt throughout the Course on a typical project to showcase the Course Learnings.

For the said purpose a Layout for an apartment structure was visualized consisting of 3 storeys.

- Ground Floor
- 1st Floor
- 2nd Floor

Each storey had 2 apartments and consisted of following features.

- 2 Bedrooms
- Kitchen
- W/Room
- Store (Single Apartment)
- Terrace (Single Apartment)

For Clarification below is the plan view of the structure under consideration for modelling.

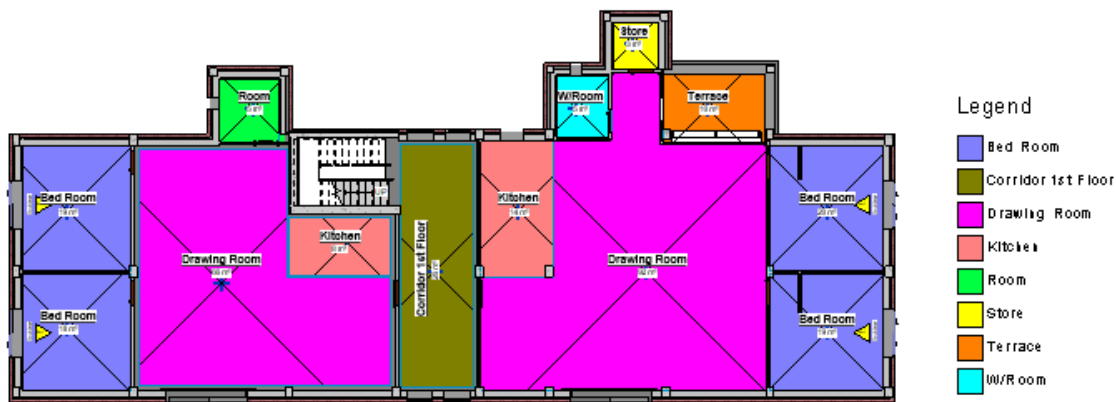


Figure 1: Typical Plan of the Building (For Each floor)

The levels were connected through a stairway enclosed within shear walls on the periphery.

The Structure Consist of typical Frame type RCC construction and Concrete blocks were used for the construction of the External/Partition and parapet walls.

The whole progress was envisaged as a collaboration between structural, Architectural and other project stakeholders. And therefore multiple models were considered as follows.

- Conceptual Model (To determine the initial/basic model of the structure for simplifying demands by an architect).

- Structural model (Linked Conceptual model overlapped as a structural engineer and detailed with frame design and Reinforcement introduction)
- Architectural model (Enriching the model with architectural elements, families, material definitions etc.)
- Final Model (Bringing the models together, removing interferences b/w structural, architectural, removing differences, and progressing towards elements, cost schedules etc.)

The whole process is stepwise followed and detailed as under.

2. Conceptual Design:

2.1. Grid and Levels

Command Grid: panel Architecture > group Datum > Grid in ribbon bar

Firstly Grid lines were made which served for positioning of the columns and the alignments of the perimetral walls. In the project browser, select the plan view and directly make the grid lines which can be locked/annotated to keep the distances rigid.

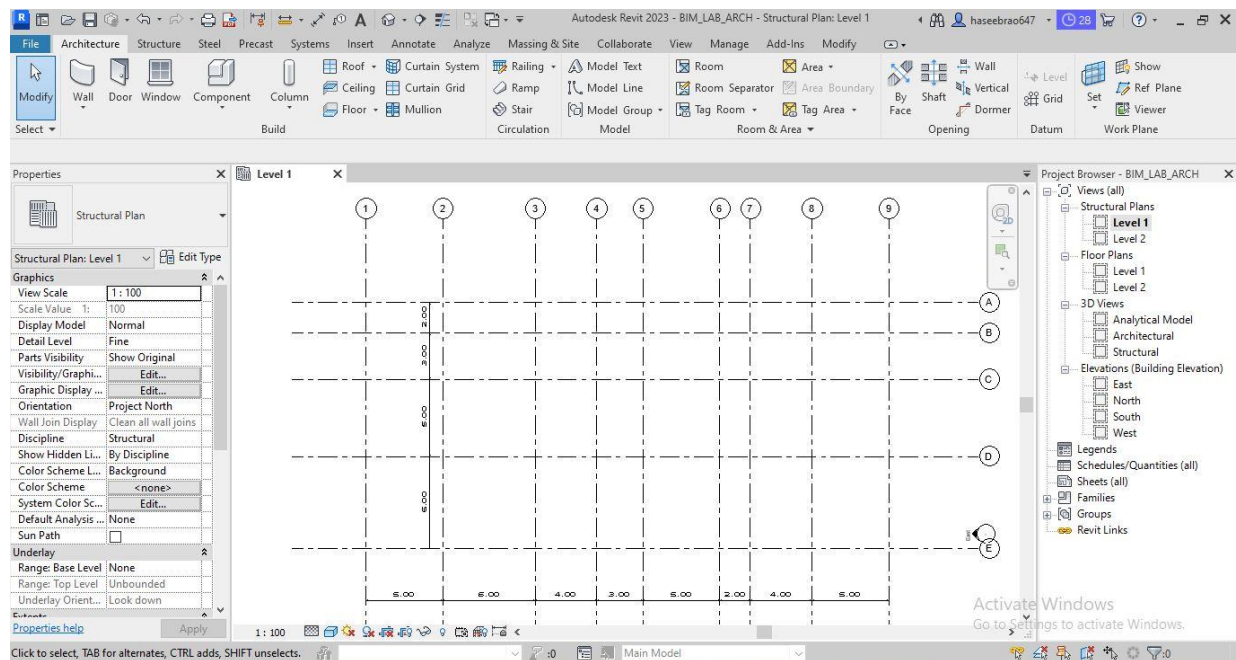


Figure 2: Grid Lines

Command Levels: Panel Architecture > group Datum > Level in ribbon bar

Levels will define the Structure in elevation (i.e. along the height). After grid lines are made the command can be followed to add levels to it.

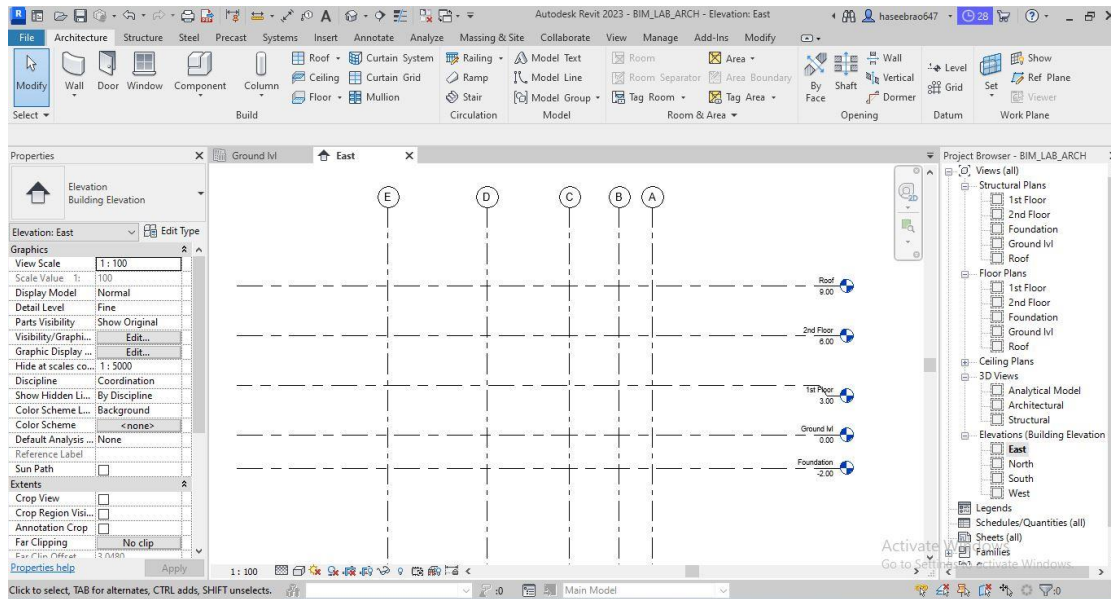


Figure 3: Levels

2.2. Object Placement

Command: panel Architecture > group Build > command Wall.

The first objects to be positioned are the perimetral walls and the interior walls, positioning in the "FFL view" (Finish Floor lvl/Ground storey). Two different walls shape were followed having different sizes for External and internal partition wall elements.

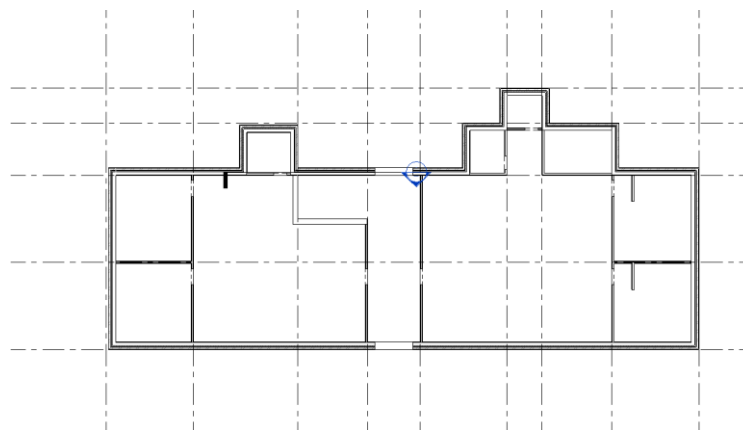


Figure 4: Wall Placement

2.3. Geometrical/Logical Constraints

Command: Lock symbol while placement or align command to align with the grid.

After placing walls, they were logically constrained to the grid lines using the lock symbol that appears when the wall is placed. Another type of constraint are the geometric ones which allow using temporary quotes, to place windows in the center of a wall. Examples are given below.

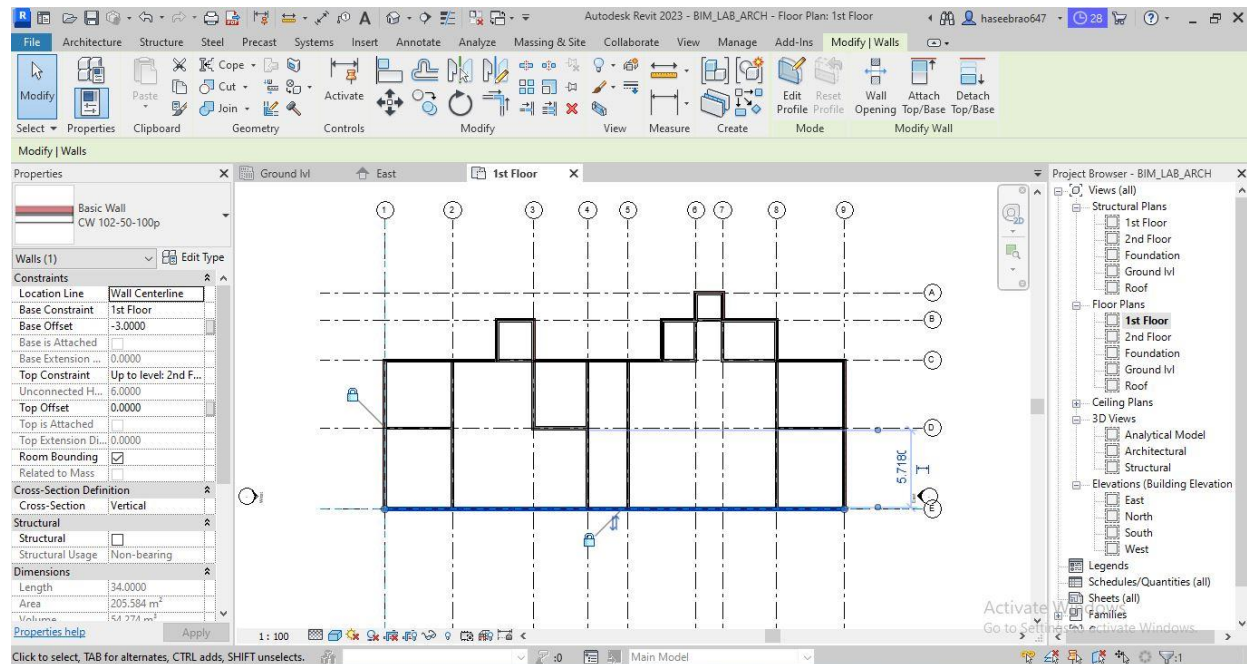


Figure 5: Constraining Walls to Grids

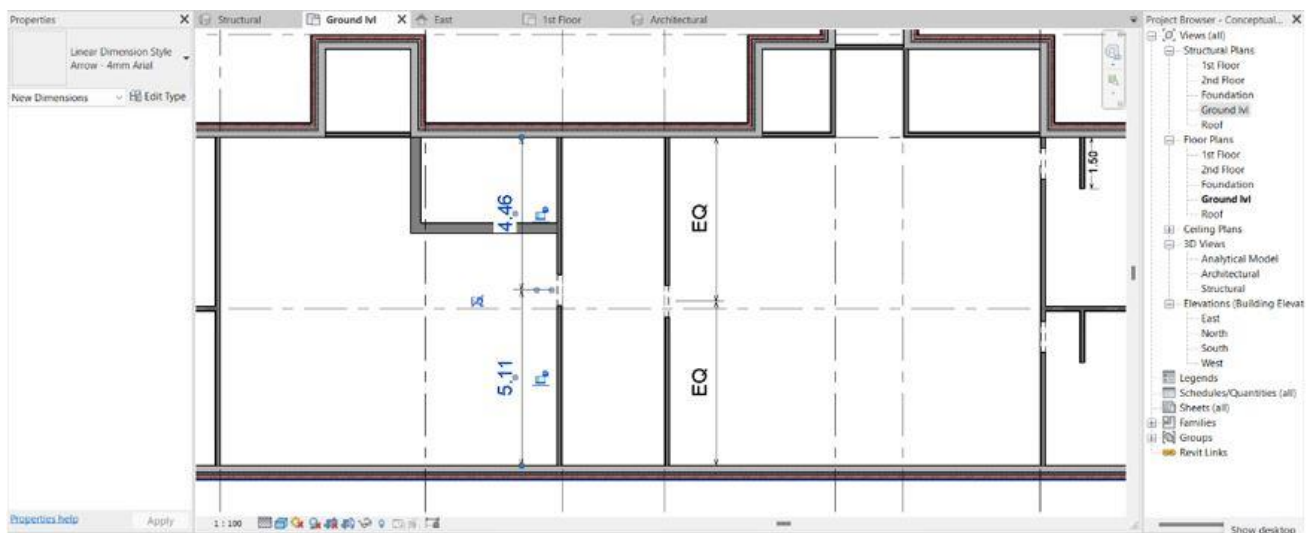


Figure 6: Annotating Objects

2.4. Parametric object properties

Command: properties > Edit Type

A number of new window and vents (for w/Room) openings were made from the properties tab. Many changes considering the height, width and sill levels etc. can be made within this tab.

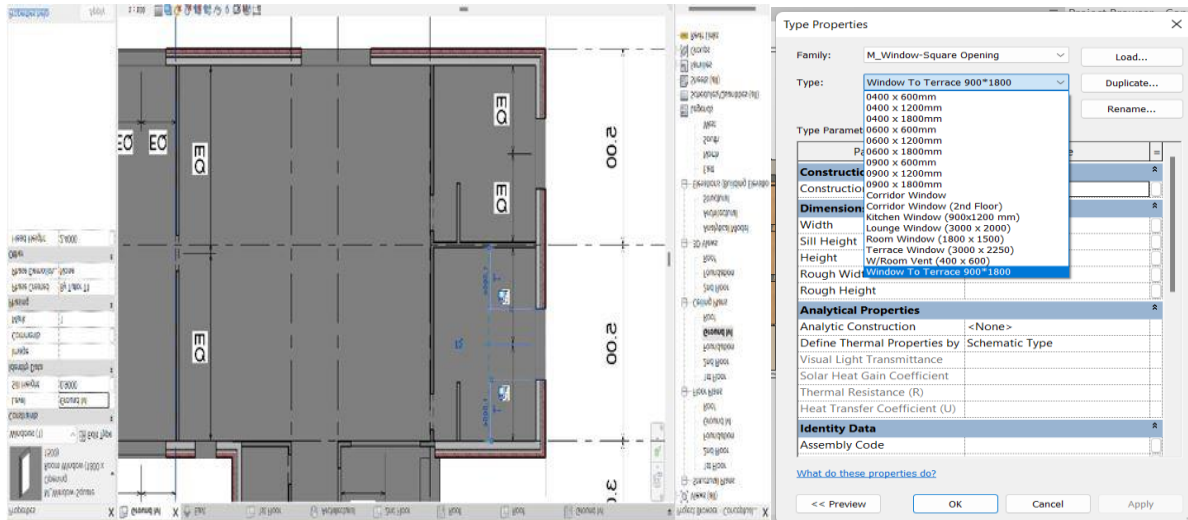


Figure 7: Placing Windows and Door Openings

While adding the windows following error appeared because the opening of windows and vents laid outside the visual presets of the plan views. For the said purpose the ranges were redefined from the properties panel. Following are the adjusted considerations.

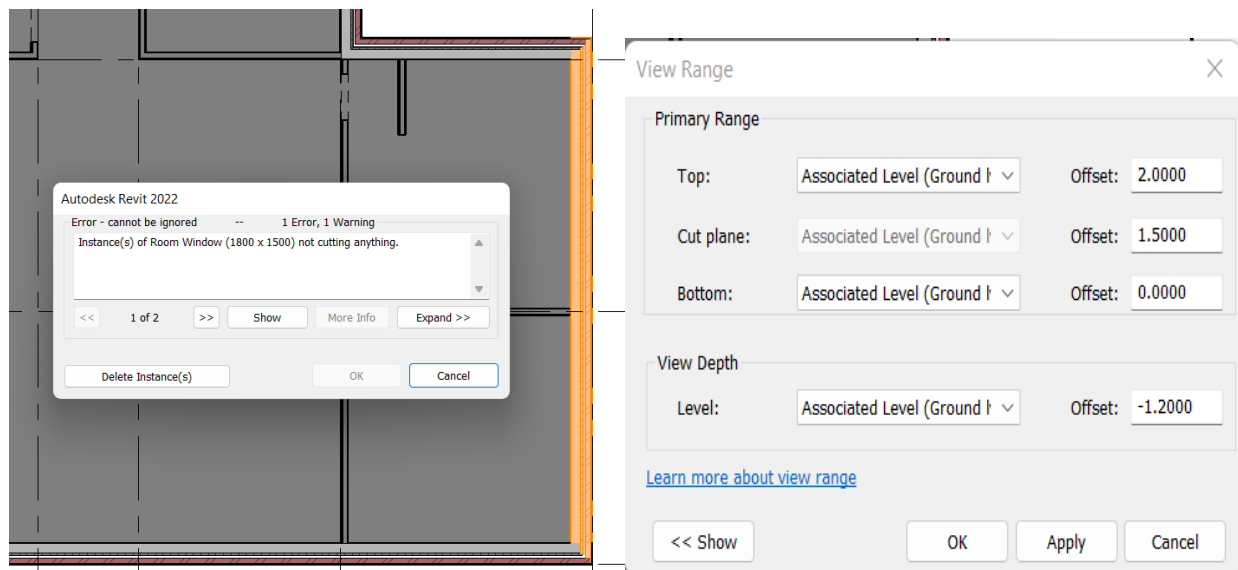


Figure 8: Adjusting View Range for Windows and Vents

2.5. Roof by Footprint

Command: panel Architecture > group Build > command Roof > sub-command Roof by footprint.

Initially 3 roofs for Ground, 1st and 2nd Stores were built with addition to a small mummy roof on top of the staircase which was essentially inclined. A “Generic - 425 mm” type was used as flooring and slab combined which was however later on redefined in the architectural and structural models.

Below image shows a typical example which was remodeled in same way for all the floors.

(Note: The roof in the case under consideration is considered to cover the walls and therefore the outer extents of the perimeter walls were considered in the footprint)

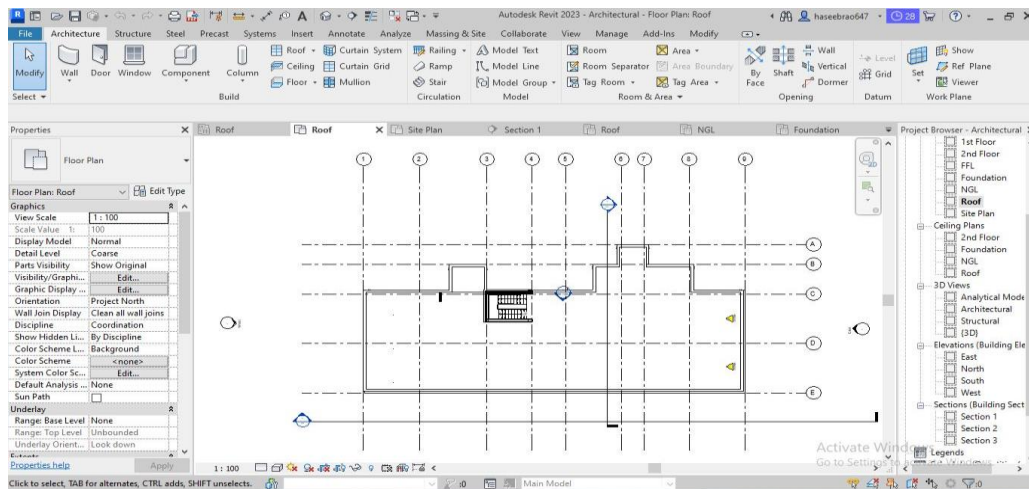


Figure 9: Upper Stores Floors by Footprint

The procedure remained similar for inclined roof with an addition of a 10 degrees pitching attributed to it from the properties.

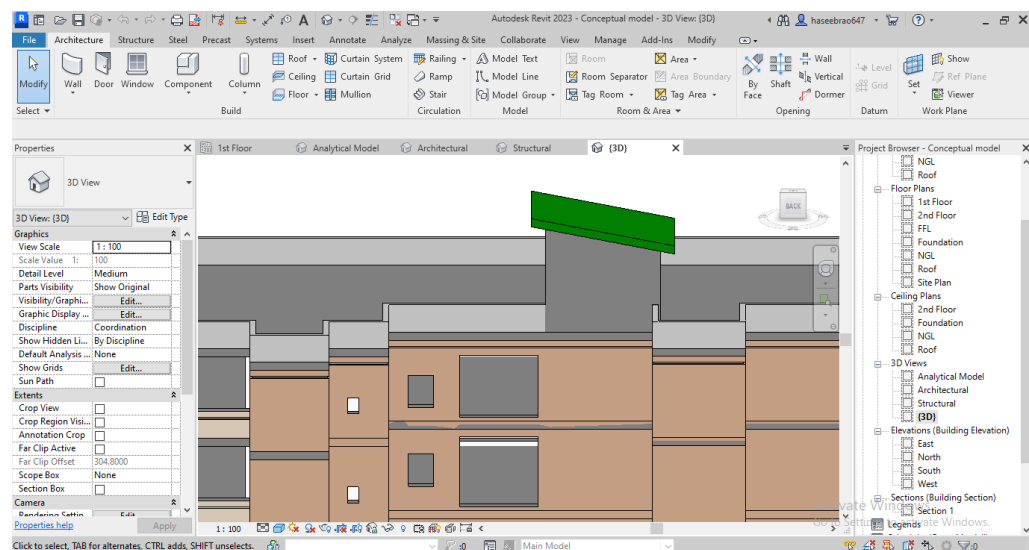


Figure 10: Inclined Roof Formation

2.6. Floor by Footprint

Command: panel Architecture > group Build > command Roof > sub-command floor by footprint.

The same process was followed but with a consideration that in this case since the floor will be resting on the ground, it doesn't need to overlap atop the walls or structural elements. Hence the internal sides of the walls were considered to makeup the floor. (For the purpose that the external walls will start atop the plinth beams, Shear walls will directly connect to foundation, (exception to partition walls as they can be built on the surface directly.)

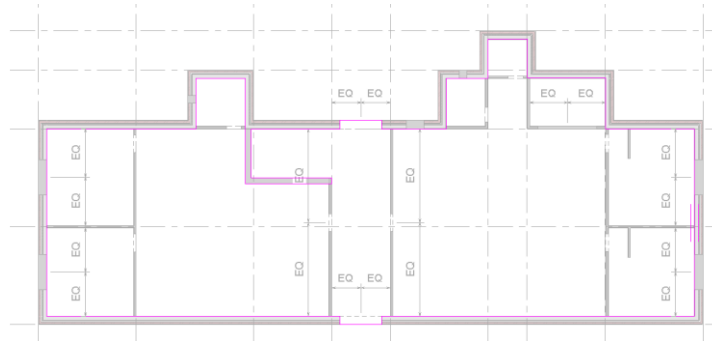


Figure 11: Ground Storey Floor by Footprint

2.7. New Type Creation

Command: Properties window > Edit Type

As a creation of a new type, A partition wall was made from duplicating a generic type wall. Properties of thickness 120mm and Metal studs were used as material however those were later changed in architectural model and Scheduling to Construction Concrete blocks.

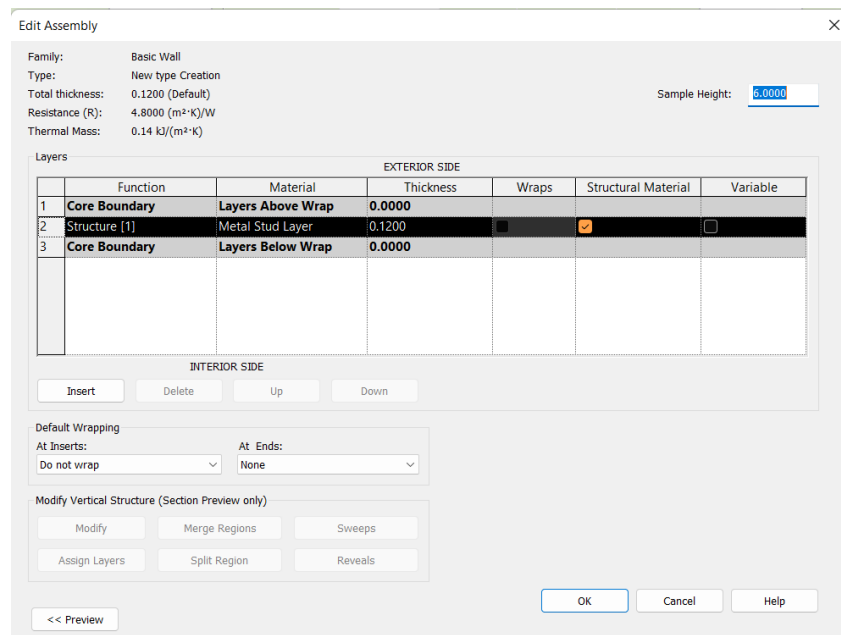


Figure 12: New Wall Type (Partition wall)

2.8. Creation of Stairs

Command: Panel Architecture > Group Circulation > command Stairs

First a shaft was made by considering the shear walls as boundaries, to cut through the floors on all levels. Afterwards for interstorey connection, stairs were made by considering the same inner boundaries of shear walls. Two runs of stairs with an intermediary platform was added in the stair plan. Once the connection was ascertained between the two storeys, the stairs were copied using the Connect levels function which directly replicates it for other floors. The cross-section view is presented here.

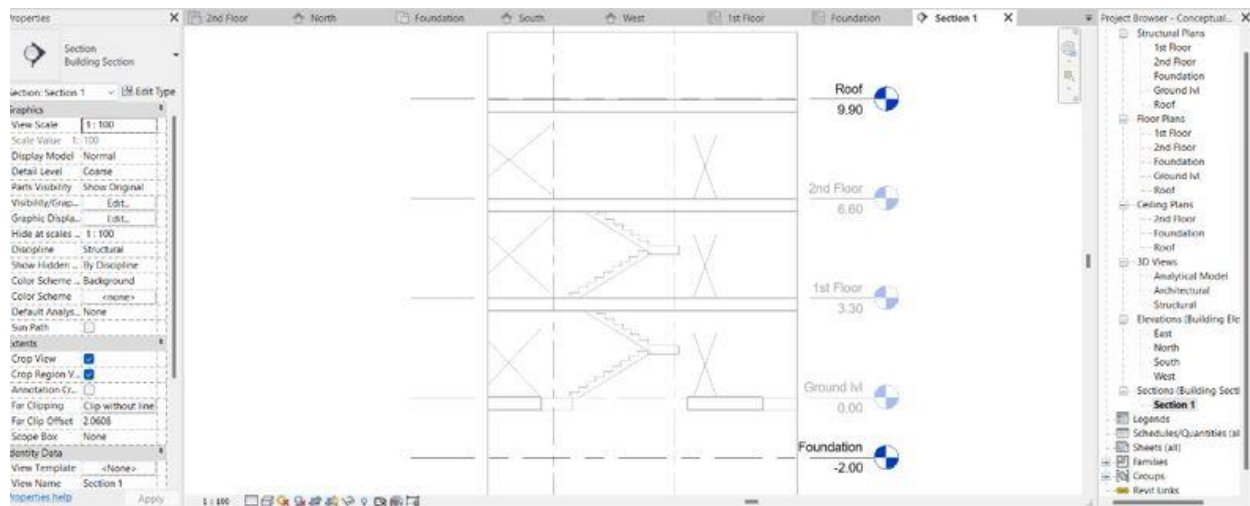


Figure 13: Connecting Levels in Stairs

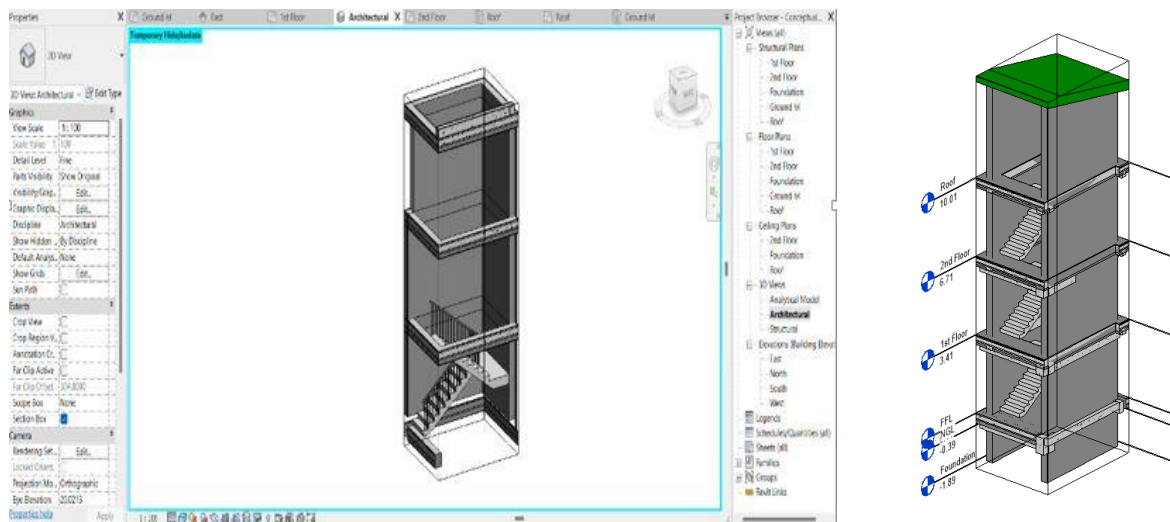


Figure 14 Stair Design

As can be seen in the below figure, the railings remained disjointed after the stairs were defined. To solve the problem from the edit Type for railings, Default join Fillet radius 100 mm, and Transition Gooseneck was selected. Results can be seen below.

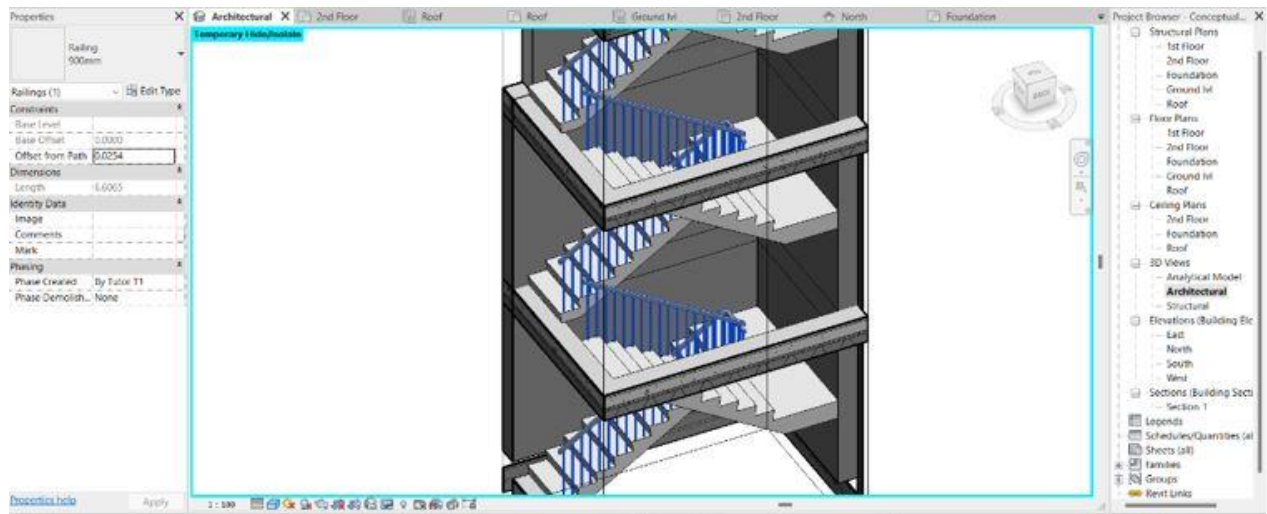


Figure 15: Breaks in Railings

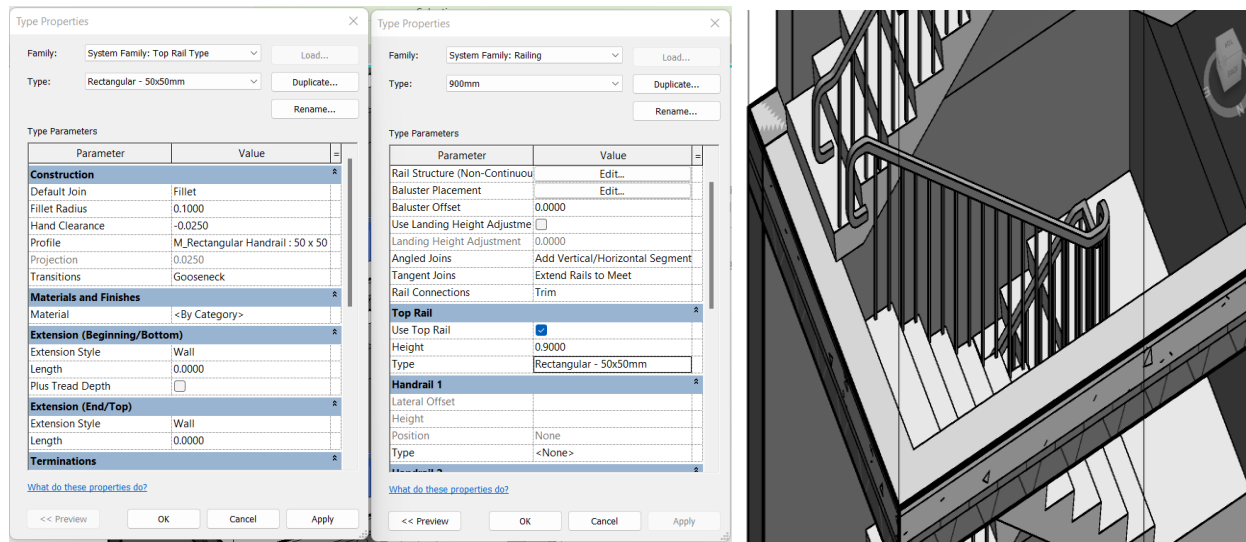


Figure 16: Adding Properties of Fillet and Goose Neck to Railings

2.9. Topography

Command: Panel Massing & Site, group Model Site > Toposurface

A rough initial topography (Later refined in Arch model) was made. For the said purpose a site plan was created and then the commands were followed to mark points to enclose the structure, material type grass was assigned to the surface.

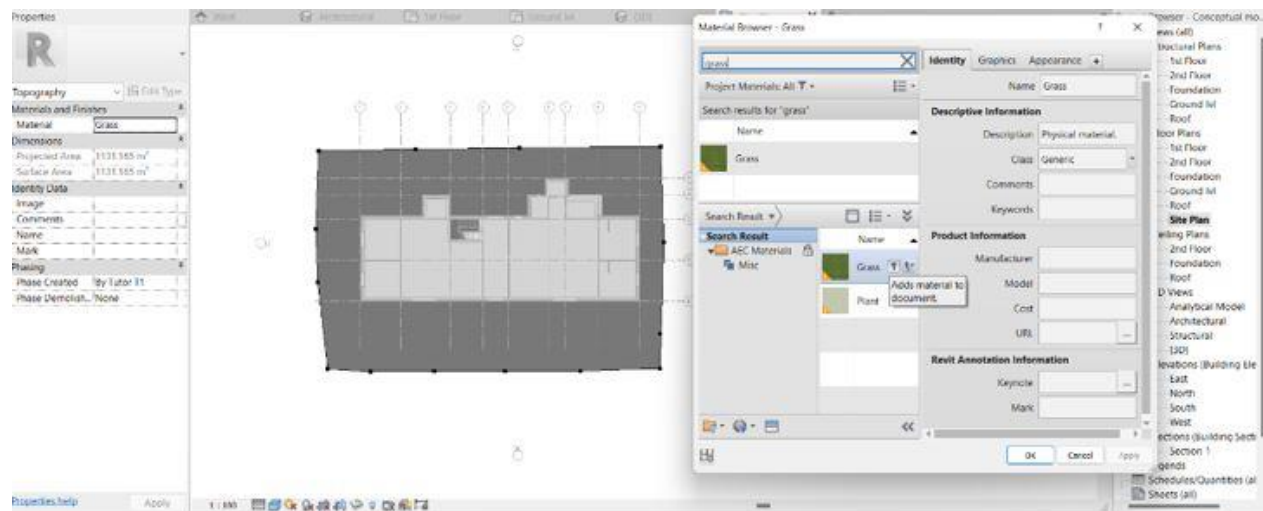


Figure 17: Rough Site Definition (Topography/Adding Grass)

3. Structural Design:

3.1. Linking Revit Conceptual model

Command: Panel Insert > Link Revit

For the structural model a new template namely BIM_LAB_STRUCT.rte was followed and the conceptual design was linked to it (Often conceptual model is prepared by architect and provided to structural engineer to start the basis of design. It serves as a tracing element for the future structure design). The conceptual model was linked through the command and internal origin to internal origin mod was followed to link the contents of Conceptual design model.

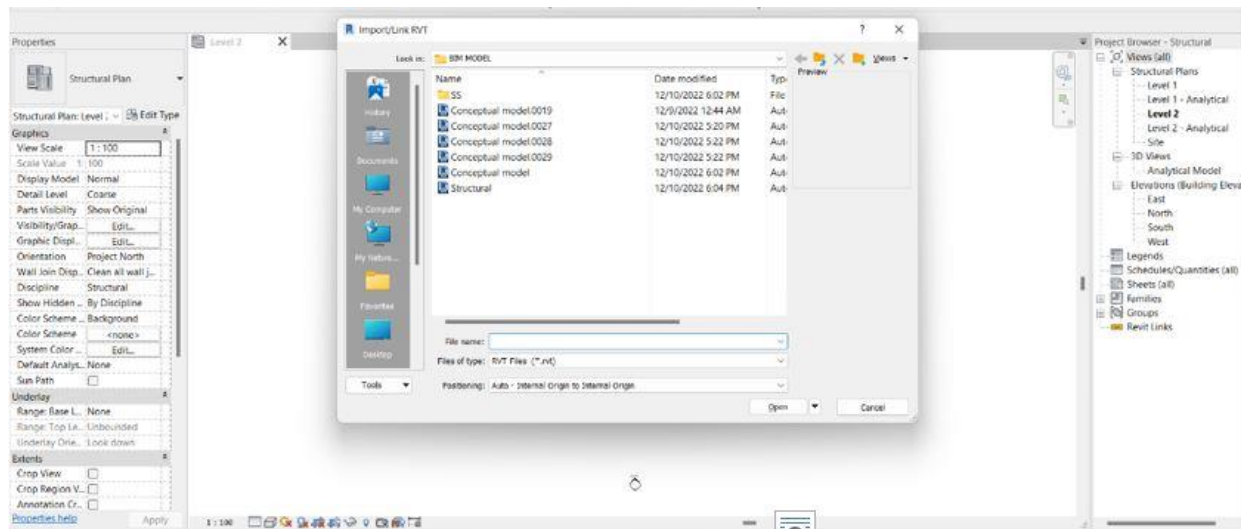


Figure 18: Linking Conceptual Model

3.2. Copy/Monitor elements

Command: Panel Collaborate > Command Copy/Monitor

Command is used to track changes in the conceptual model at a later stage and import some grid and level details etc. which can be directly cross referenced and commented upon.

After entering the Copy/Monitor function, by clicking on the command Copy the grids and levels were selected. An ECG icon appears to confirm the copying.

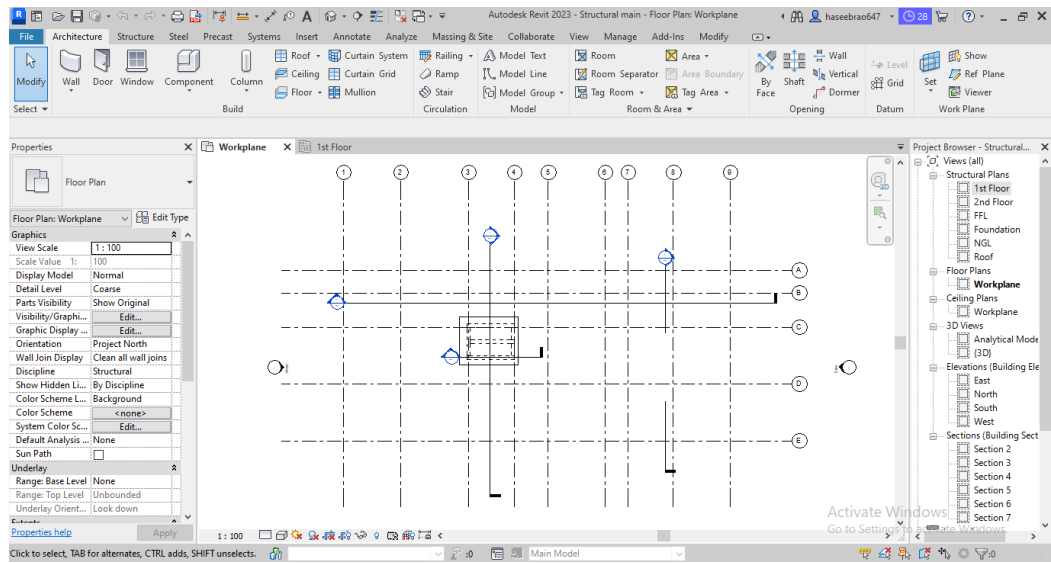


Figure 19: Copy/Monitor Grids

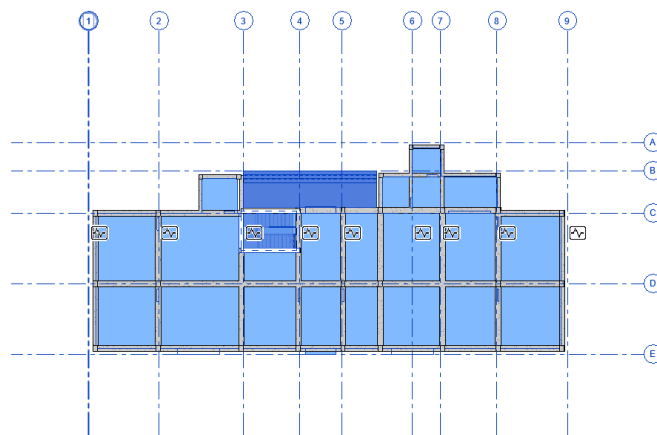


Figure 20: Grid Monitoring

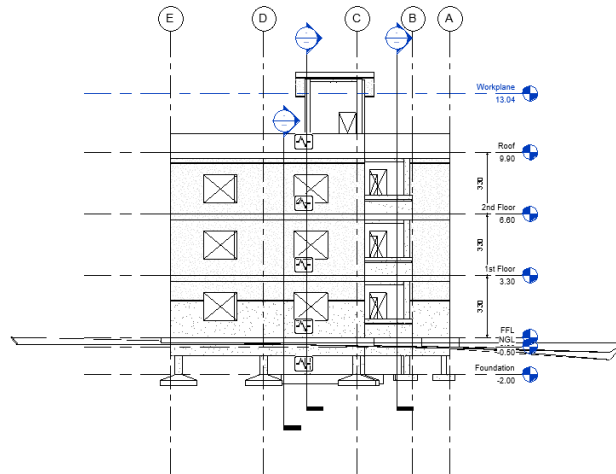


Figure 21: Level Monitoring

3.3. Changes Detection

Command: Panel Collaborate > Coordination review

To have a better concept of how review works an artificial mistake was introduced in the names of the grid lines and later corrected. The change was detected, and rivet asked for a Coordination review. Therefore following the command within the panel, a number of decisions like acceptance, rejection or commenting for review can be done.

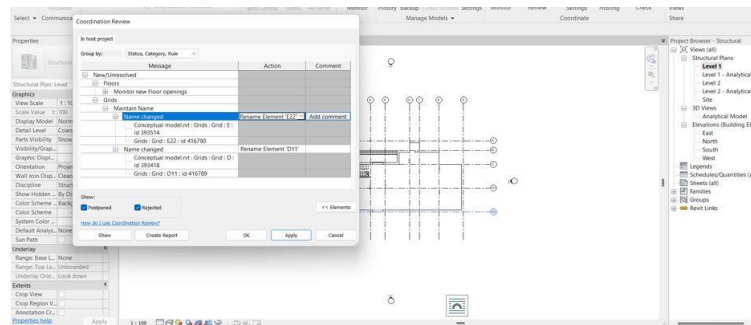


Figure 22: Change Detection (Co-ordination Review)

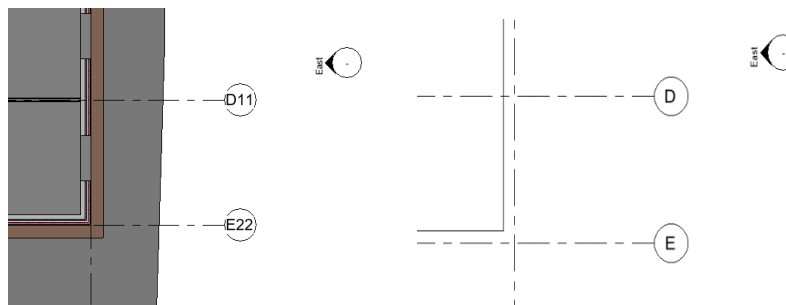


Figure 23: Changed Grid Names

3.4. Structural Elements

3.4.1. Structural Columns:

Command: Panel Structure > Group structure > Command Columns

Firstly, Structural columns were placed at respective positions in the structure. Two types of columns were designed,

- Extended Edges Columns 0.3 m x 0.3 m (Extended from regular rectangular shaped slab)
- Main Structural Columns 0.3 m x 0.4 m (Main Slab Columns)

The columns were placed at respective positions in the structure afterwards.

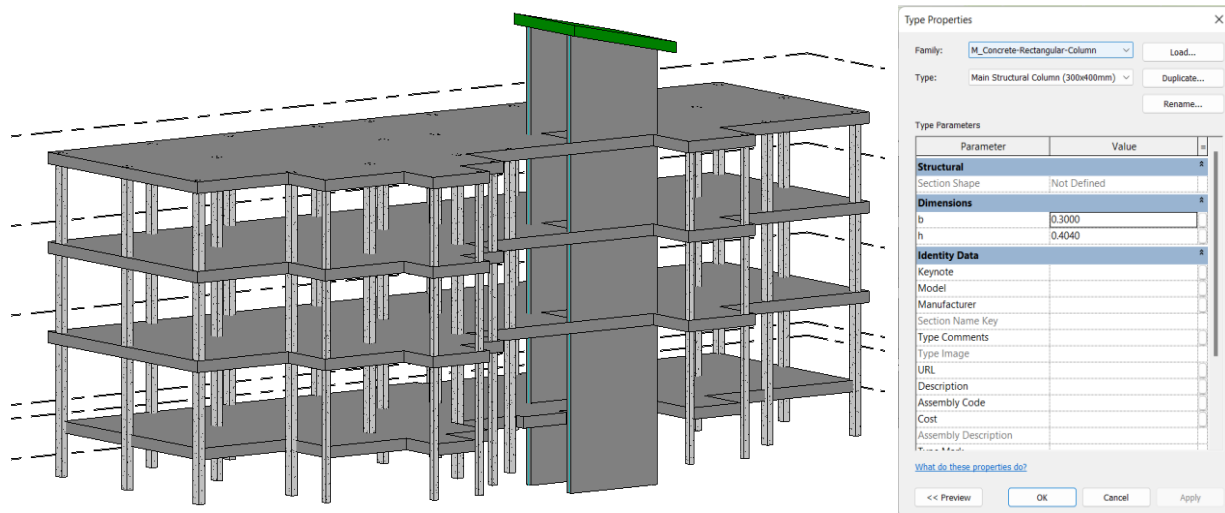


Figure 24: Structural Columns

3.4.2. Structural Beams

Command: Panel Structure > Group structure > Command Beams

Afterwards to Connect the Columns structural beams were made to complete the frame structure using the command. Following were the type of beams made.

- Main Floor Beams 0.4 m x 0.7m (Includes floor and Plinth Beam under main structure)
- Beams Extended region 0.25m x 0.5 m (Includes extended region beams including plinth beams)
- Mummy floor beams 0.25m x 0.5m (under slab of Roof/mummy floor, same beam because less load.)
- Inclined roof beams 0.3m x 0.3m (Present on the inclined slab atop Stairs shaft)

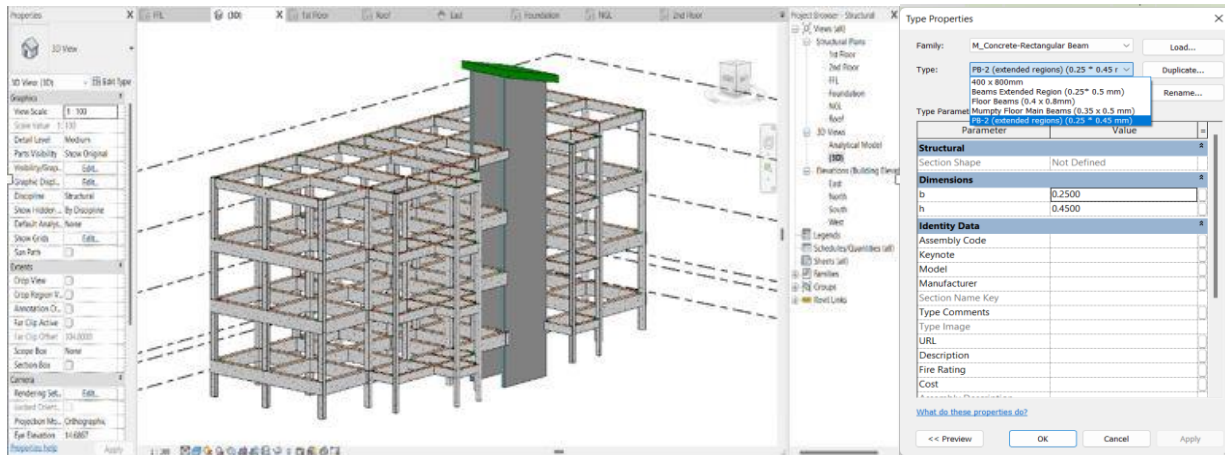


Figure 25: Primary/Secondary Beam Meshes

3.4.3. Beam Systems

Command: Panel Structure > group Structure > command Beam System

As for beam systems only single secondary Beams (0.3 m x 0.4 m) in between the center making a cross between the main Beams were drawn as can be seen in the following figures.

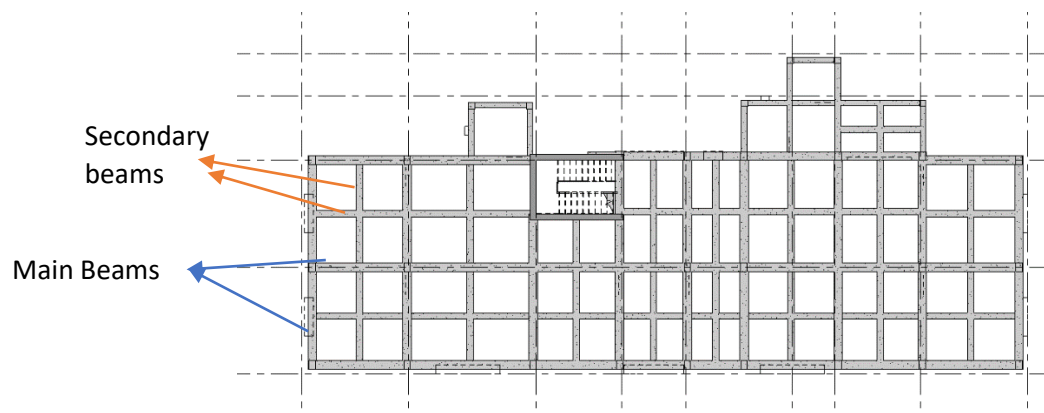


Figure 26: Automatic Beam System

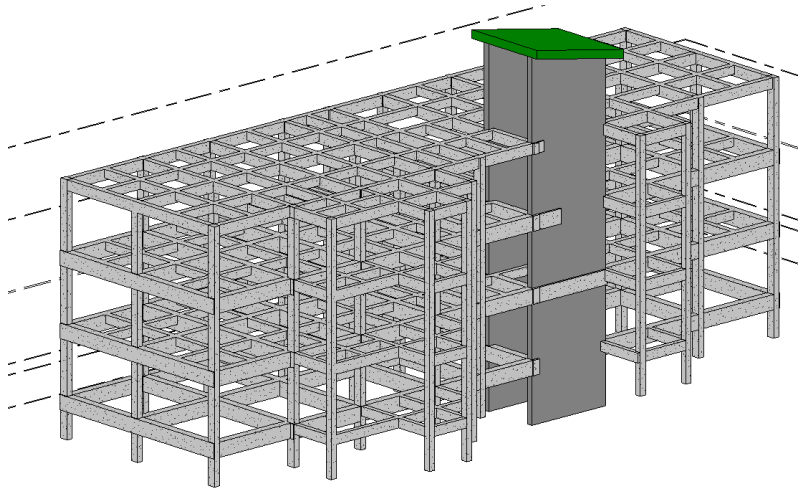


Figure 27: Beam Systems (3D-View)

3.4.4. Structural Foundation

Command: Panel Structure > group foundation > command isolated

The foundation for the extended edged columns were made with following considerations.

Square Foundation Column (LxWxH) = 1.2 m x 1.2 m x 0.3 m.

The foundation for the other columns were specialized extruded design elements explained further on.

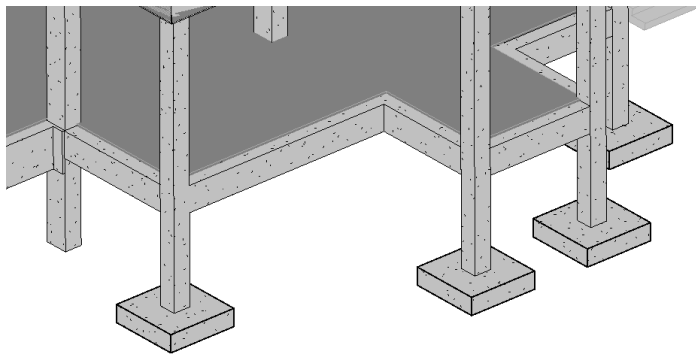


Figure 28: Simple Square Foundation (Columns External)

A slab type foundation 5.5m x 7 m with 0.5m thickness was drawn to support the shear walls of the stair case. The command line was similar to above but the command used was slab instead of isolated.

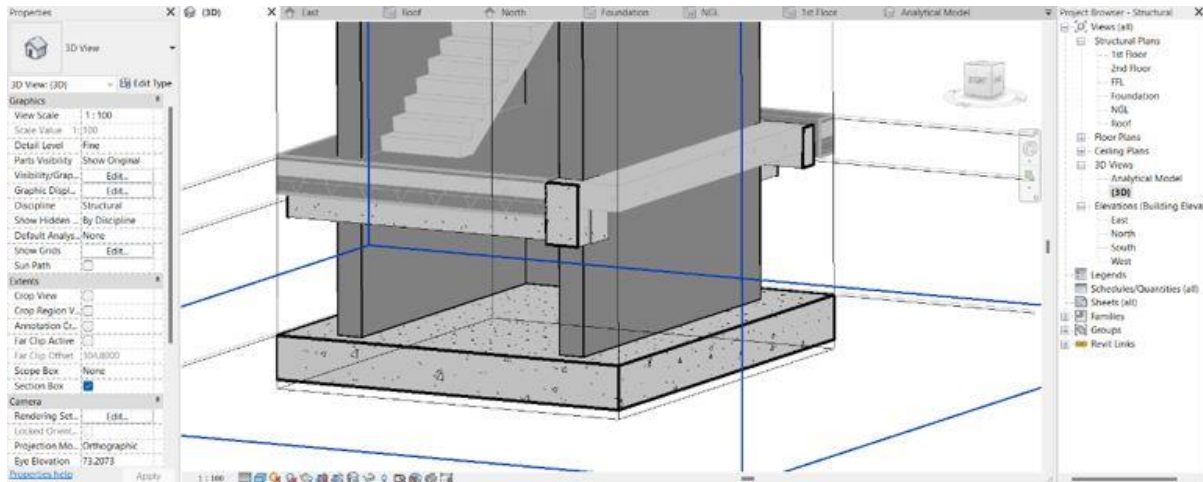


Figure 29: Mat Foundation for shear walls of stairs

3.4.5. Structural Slabs

Command: Panel Structure > group Structure > Command Beam System

Structural slabs were envisaged the same way as in previous model by superimposing on the conceptual linked model.

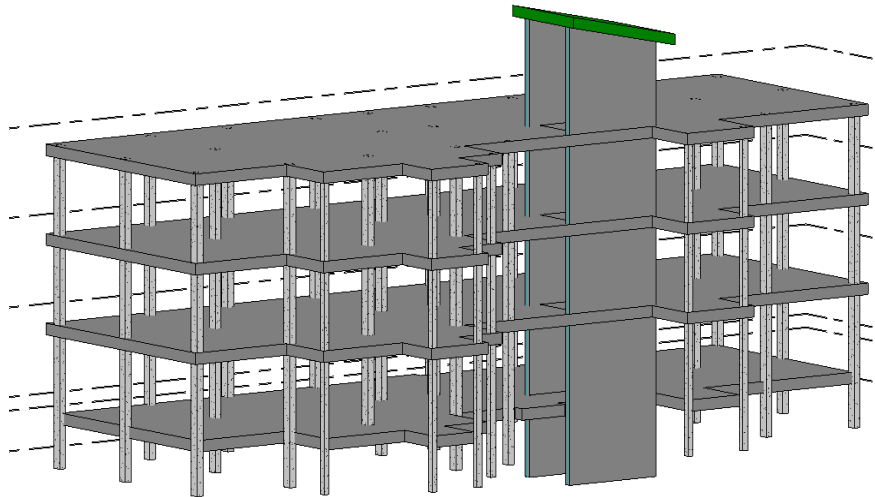


Figure 30: Structural Slabs

3.5. Objects Layering

Command: Properties window > Edit type

After selecting the structural roof and opening the Edit Type window certain properties can be modified such as material definition and associated thickness. Also the importance of material as an insulation, structure or finishes etc. can be ascertained. A sample Concrete slab to define how the object layering works is shown here. This was later modified to concrete slab with screed finish as structural slab and for architectural some floorings were added depending upon the floor usage.

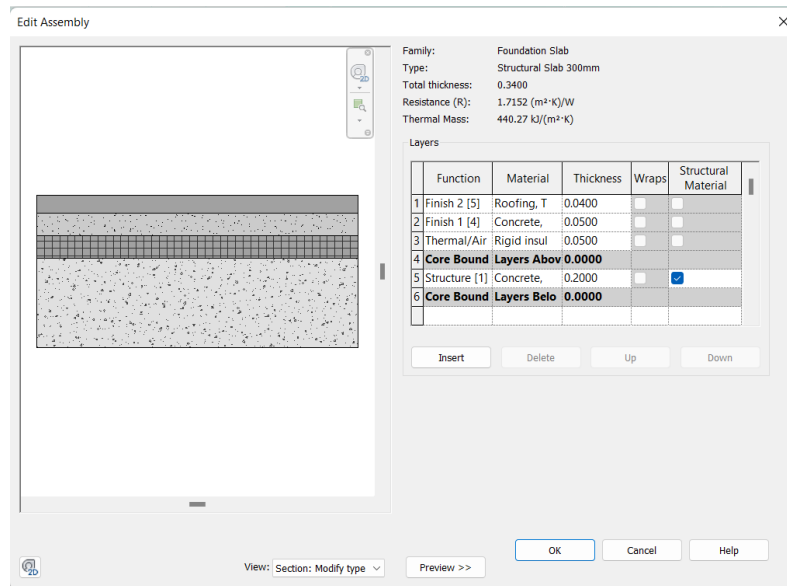


Figure 31: Sample Slab layering

3.6. Work plane definition

Command: Panel Structure > group Work Plane > command Set

The process was followed to detail the work plane definition in case of inclined roof with beams which although isn't normally the case in shear wall-Slab type connections.

In rivet normally inclined slab is easier to place compared to inclined beams. Extra process of selecting an inclined plane and then model lines have to be followed.

The lower face of the conceptual was selected after checking the "Pick a Plane" radio button.

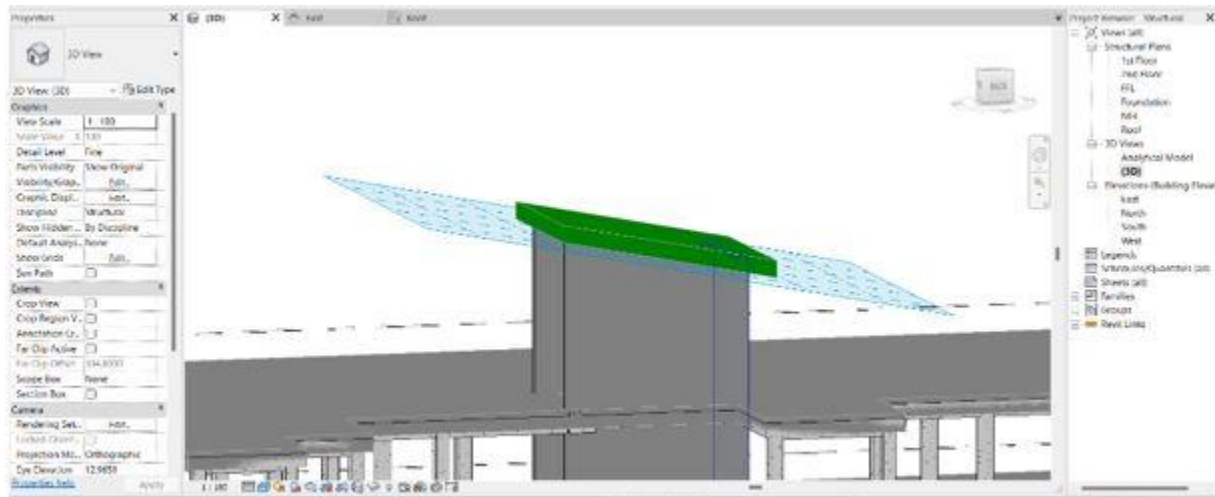


Figure 32: Workplane Definition for Pitched Roof on stairs

3.7. Model Line Exploitation

Command: Panel Structure > group Model > command Model Lines

The structural beams were located in the correct position by taking as a reference the model lines that were drawn to define the boundary of the bottom part of the roof and the axis of the beams.

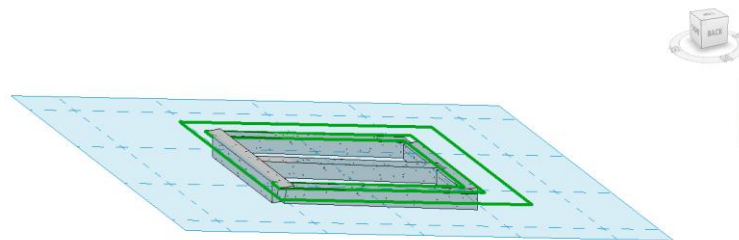


Figure 33: Creating Model Lines (Increased Thickness)

3.8. Model in place objects

Command: Panel Structure > group Model > command Component > sub-command Model in Place

An isolated trapezoidal footing was chosen for the Main Structural columns to be designed as part of the model in place object. In foundation level plan view, Base foundation shape of 2 m x 1.5 m extruding it to 0.3m after which blend command was used. The top surface was the respective main column size. The thickness for the blend was also 0.3m which makes the total thickness from the top face to the bottom face of foundation of about 0.6m in total.

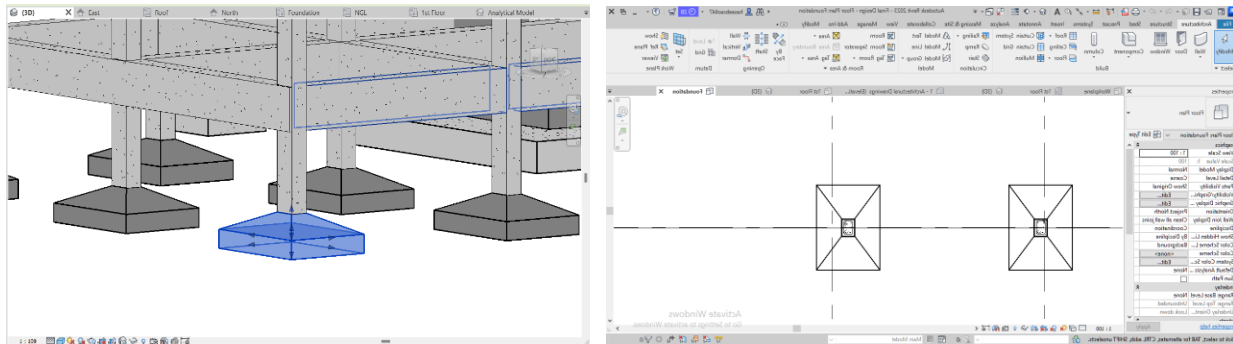


Figure 34: Modelling in-place objects /Extrusion (Trapezoidal Foundation)

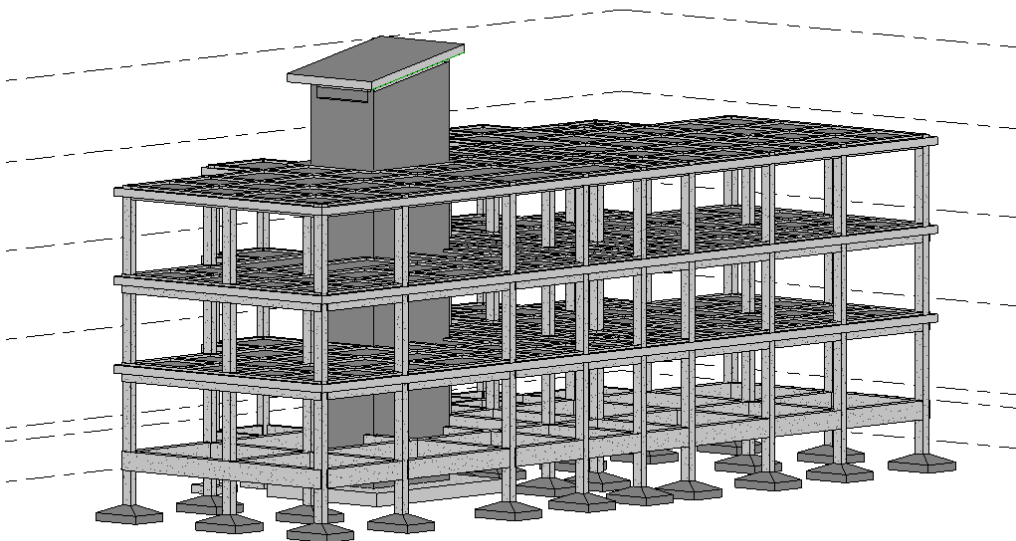


Figure 35: Modelling in-place objects /Extrusion (Trapezoidal Extended)

3.9. Reinforcement Design

After the creation of the structural elements, reinforcement was to be added in each individual element. As opposed to a sample element be enriched with the reinforcement whole project was reinforced and copying feature was exploited.

The reinforcement remains hidden in certain views and can be individually shown by modifying the visibility states from the properties window.

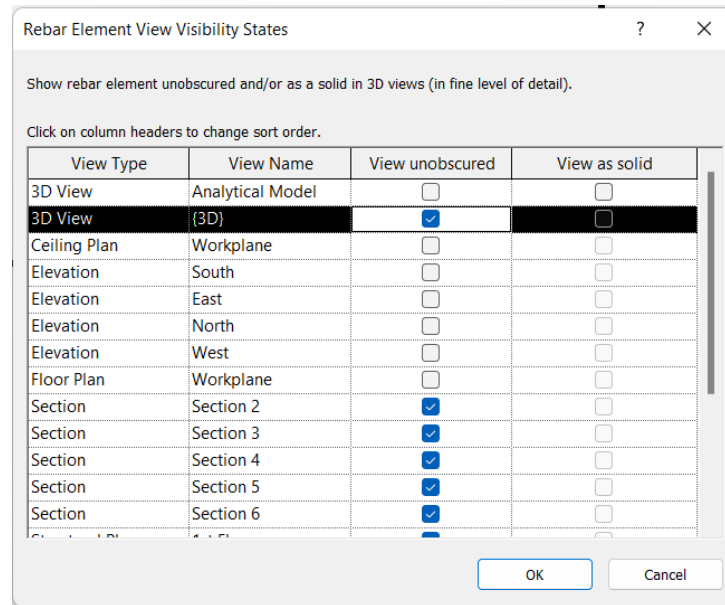


Figure 36: Adjusting visibility states of reinforcement

3.9.1. Column Reinforcement:

Command: Select Element > Rebar

The total project was divided into two columns, Main structural and extended region columns.

3.9.1.1. Longitudinal Reinforcement

To position the longitudinal bars, the “Current Plane” was used as the Placement Plane and the “Parallel to Cover” as Placement Orientation.

Column type	Dimensions (m x m)	Type of Bars	Diameter	No. of bars
Main structural Column	0.3 x 0.3	Rebar shape 1 *	22 M	6
Extended Region Column	0.3 x 0.4	Rebar shape 1 *	22 M	4

*Rebar Shape 1 = Dowel at one end and straight at other (L shaped)

Type of bars, were the M_00 with a diameter of 13mm are used, 4 bars are positioned in the corners of the column.

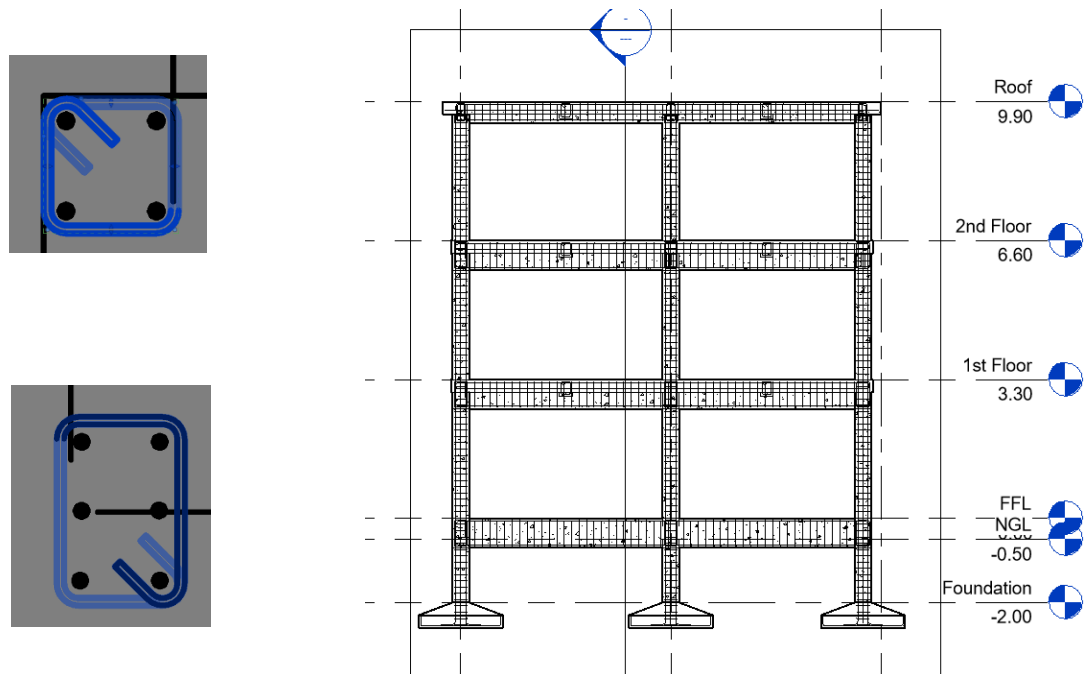


Figure 37: Column Reinforcement Detailing

3.9.1.2. Transversal Reinforcements

For the positioning of the stirrups, after selecting the “Current plane” is used as a placement plane and “Perpendicular to cover” as placement orientation. For both the columns the type of stirrups chosen were M_T1. Once the first element was positioned, the spacing was adapted.

Column type	Dimensions (m x m)	Type of Bars	Diameter	Spacing
Main structural Column	0.3 x 0.3	M_T1	13 M	150 mm
Extended Region Column	0.3 x 0.4	M_T1	13 M	200 mm

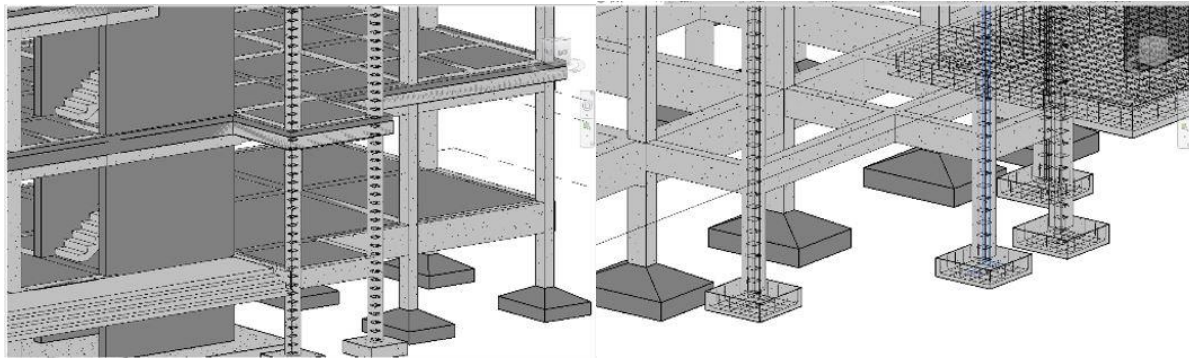


Figure 38: Transversal Reinforcement (Stirrups in Columns)

3.9.2. Slab Reinforcement (Area)

Command: Select element > group rebar > command Area

Once the desired element has been selected, the command Area, in group Rebar, is used. Whole area of slab is selected and after the steel grid positioned it is possible to modify the spacing between the bars, their diameter and the type of hook in the terminal end.

In this case a single mesh of bottom bars with following detailing were added as part of the slab structure.

Element	Type of Bars	Diameter	Spacing
Main Structural Slab	M_00	13 M	250 mm

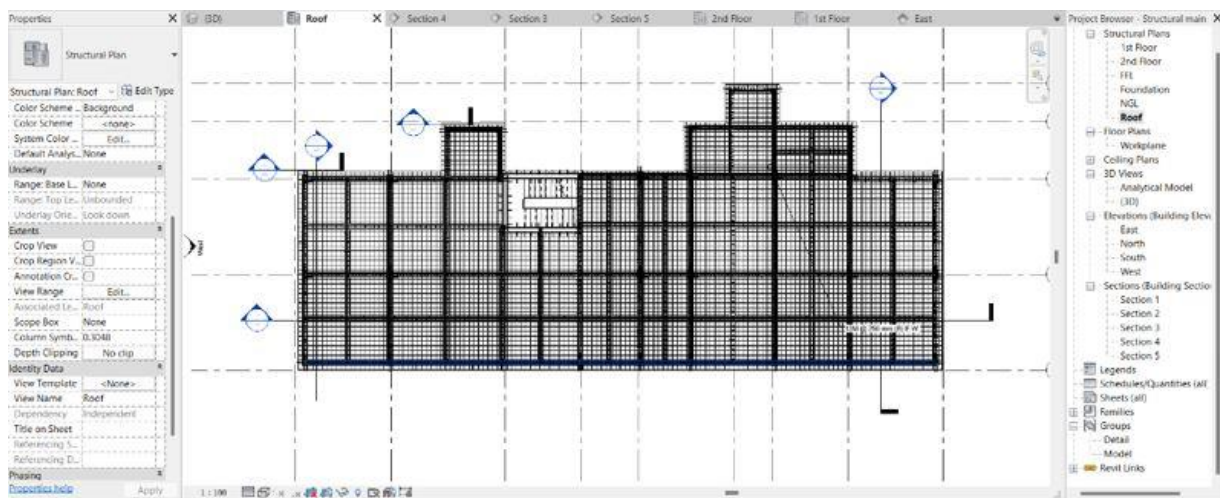


Figure 39: Structural Reinforcement plan (Slab)

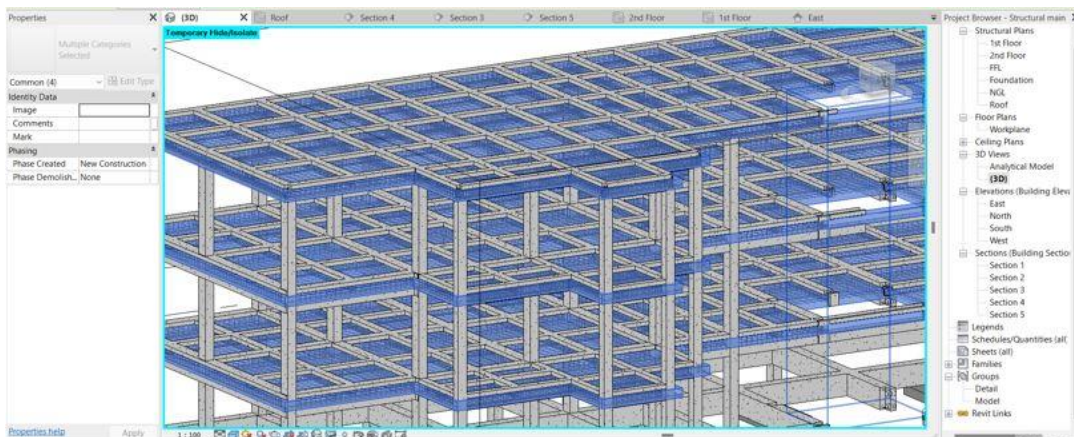


Figure 40: Slab Reinforcements (3D)

3.9.3. Beam Reinforcement

Command: Select Element > Rebar

For the beams the process is similar as of columns with an exception that Transverse cross-section have to be in view for the beam, to locate both longitudinal and transverse reinforcements. For this purpose, with the section command in the panel View, group Create a section has been placed.

Beam type	Dimensions (m x m)	Type of Bars	Longitudinal Bars		Transverse Bars	
			Bottom	Top	Type of Bars	Spacing
Roof (Mumpty Floor Beams)	0.30 x 0.50	M_01	3 ϕ 22M Bars	2 ϕ 22M Bars	M_T1	Φ 10M 150 mm
Main Floor Beams + Plinth Beams	0.40 x 0.70	M_01	4 ϕ 22M Bars	3 ϕ 22M Bars	M_T1	Φ 13M 180 mm
Secondary Beams	0.30 x 0.40	M_01	3 ϕ 16M Bars	2 ϕ 16M Bars	M_T1	Φ 10M 250 mm
Beams Extended Region	0.25 x 0.50	M_01	3 ϕ 22M Bars	2 ϕ 22M Bars	M_T1	Φ 13M 180 mm

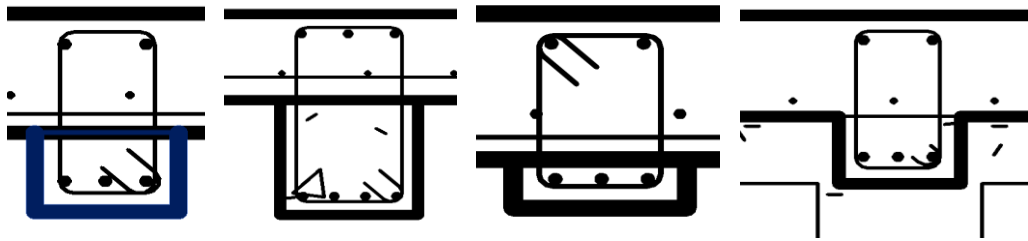


Figure 41: From Left to Right a. Roof Beams, Main Floor Beams+Plinth Beams, Secondary Beams, Beams Extended Region

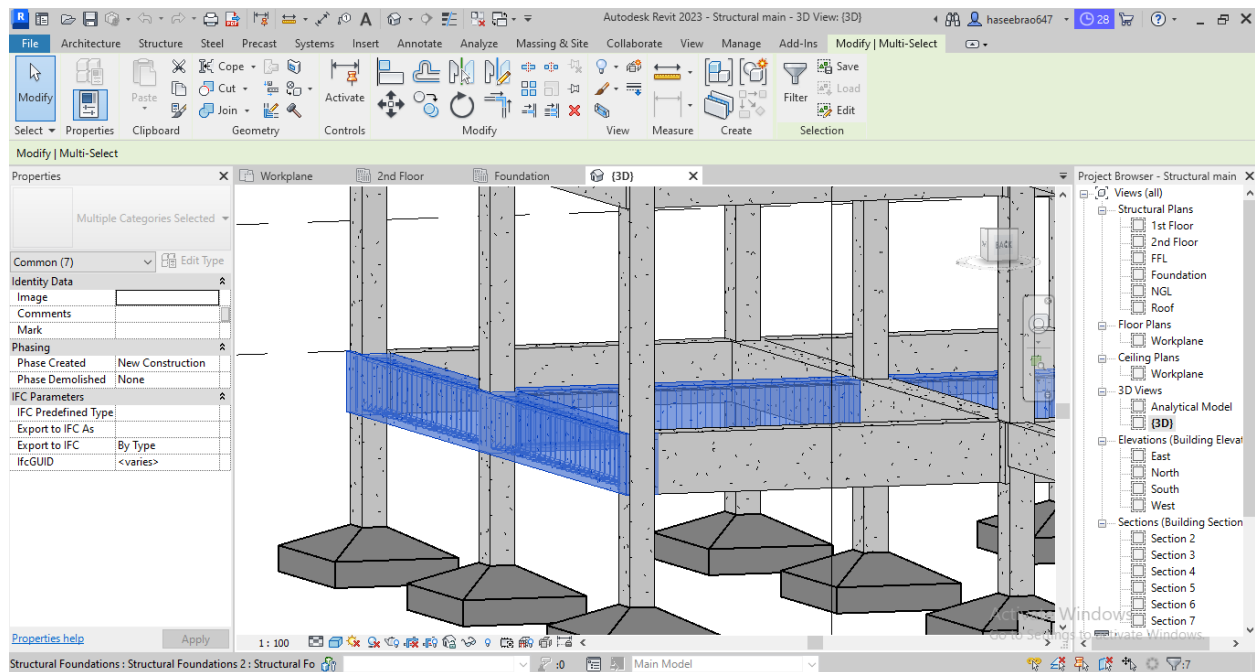


Figure 42: Reinforcement definition in Beams

3.9.4. Shear Wall Reinforcement

Command: Select element > group rebar > command Area

once the desired element has been selected, the command Area, in group Rebar, is used. Whole area of shear wall is selected (inclined at top end) and after the steel grid positioned it is possible to modify the spacing between the bars, their diameter and the type of hook in the terminal end.

In this case a single mesh of bottom bars with following detailing were added as part of the slab structure.

Element	Type of Bars	Diameter	Spacing
Shear Walls	M_02	16 M	400 mm

The shear wall has two meshes on both faces of the walls.

For the horizontal bars 90° hooks at both ends were introduced through properties window for both meshes.

For vertical bars dowels at bottom end and hooks inserted in top slab were made. Manual correction at top end with connection to inclined slab were made for few bars because they were not covered in concrete completely.

3.9.5. Foundation Reinforcement

Command: Select Element > Rebar

For all foundations 16M bars @ 250 mm spacing were introduced in both directions with 90 degrees hooks at both ends using the command above stated.

Element	Type of Bars	Diameter	Spacing
Shear Walls	M_02	16 M	250 mm

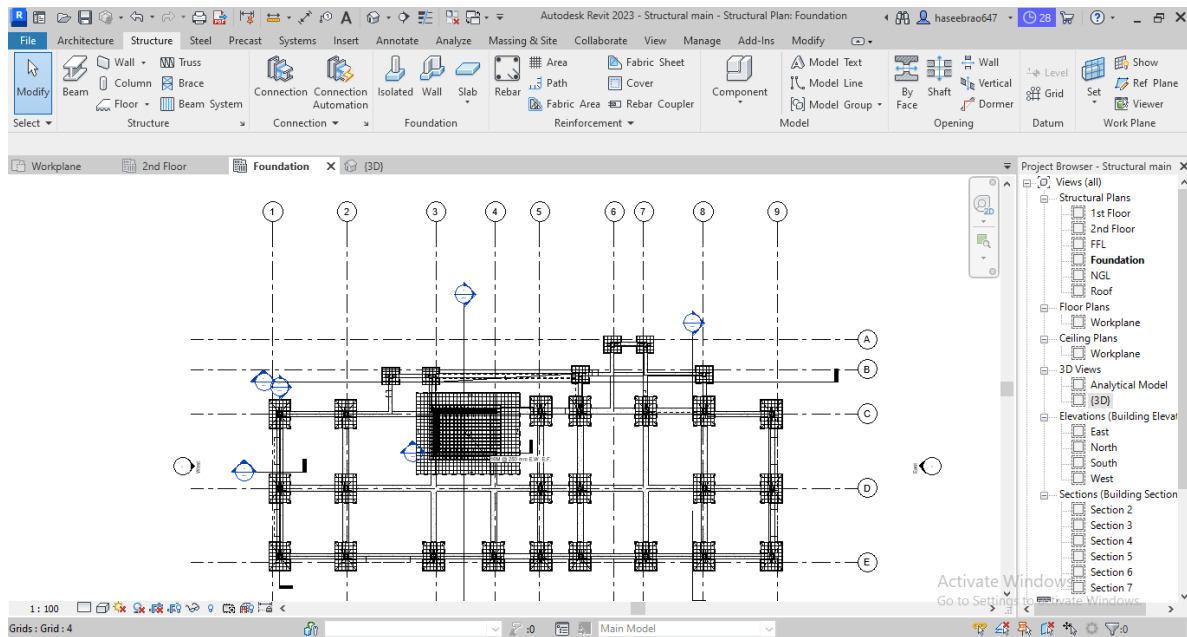


Figure 45: Foundation Reinforcement (Structural plan)

Final Reinforcement 3D dense meshing is shown as under,

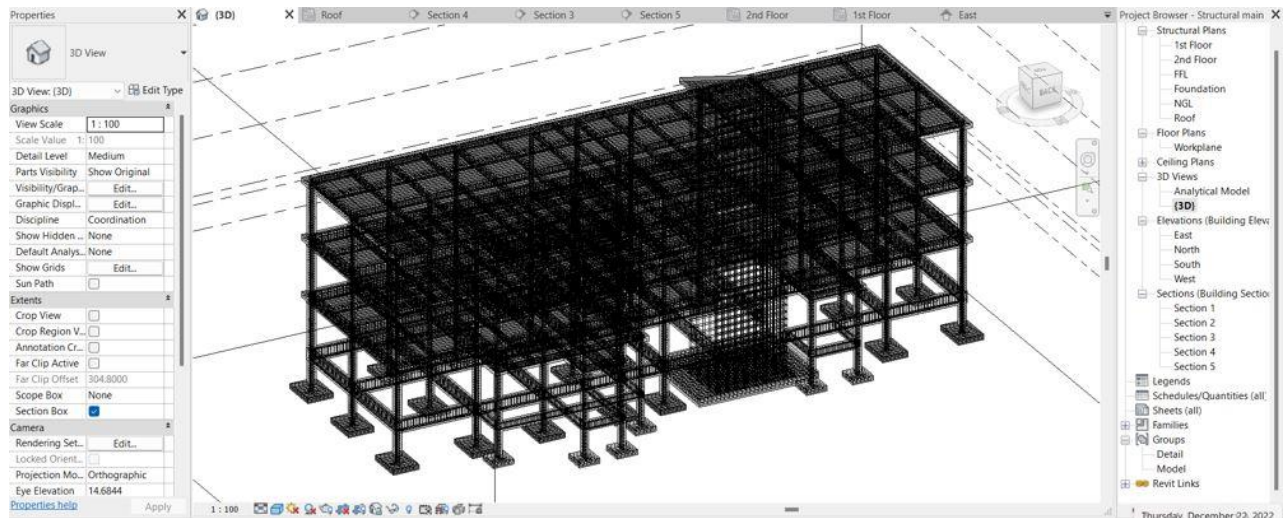


Figure 46: Final Reinforcement Meshes (3D)

3.10. Rebar Cover Setup

Command: Panel Structure, group Reinforcement, subgroup Rebar Cover Setup

To add manual rebar cover setup for structural elements, the command can be used to define the covers directly to the respective elements. Four new rebar covers were added For Slabs, Beams, Columns, Foundation (30,40,30,50 mm) respectively.

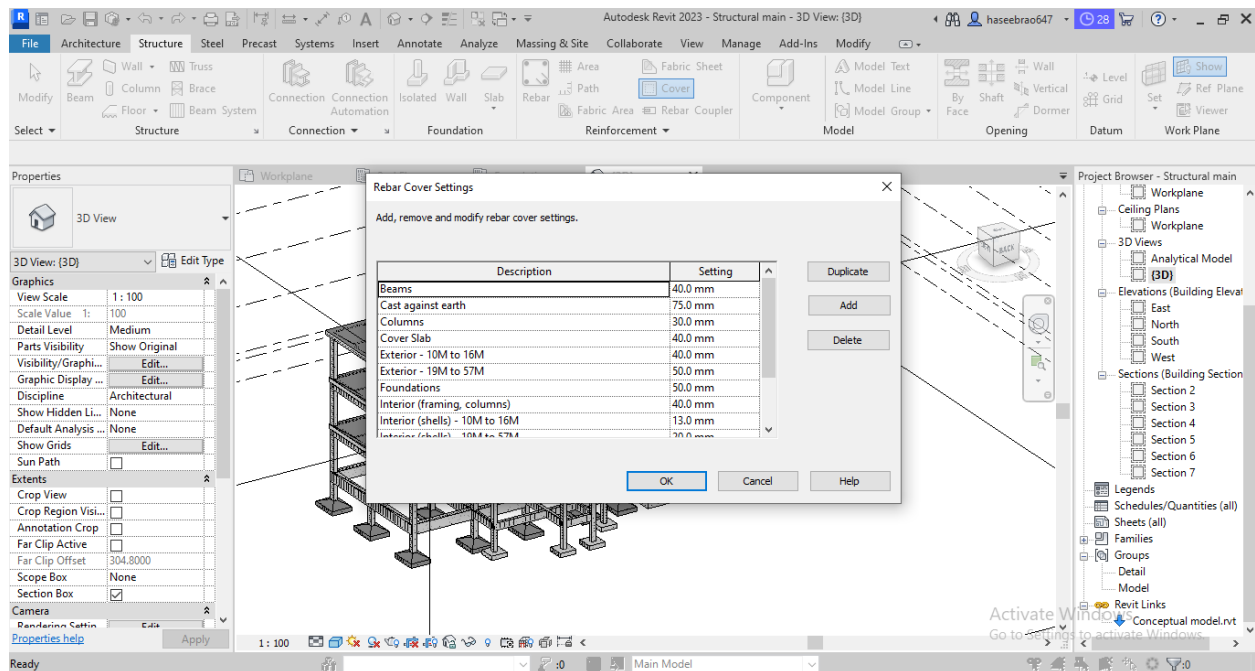


Figure 47: Defining New Concrete Covers

3.11. Analytical model

One of the multi useful features of Rivet in the analytical model formation of the structure. It's build directly with the creation of the structure and can be directly analyzed or introduced to certain structural analysis software's to check for the admissibility and adequate designing.



It can be opened directly from the bottom ribbon by clicking the highlighted button. If there is a disconnect between members they can be directly corrected by moving them in 3 dimensions.

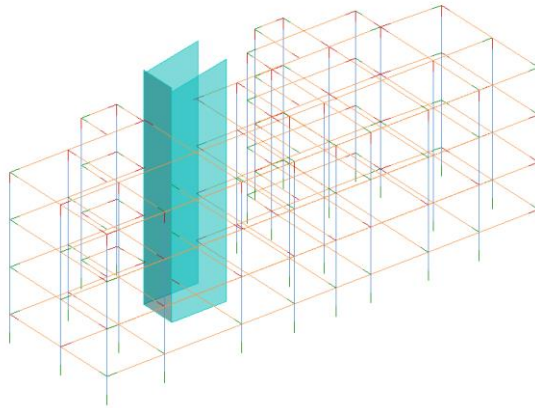


Figure 48: Analytical Model Complete

A supplemental override was performed on our part by assigning certain filters and overrides for connected and unconnected analytical nodes of the model by going into visibility/graphics over-rides > Filters.

The filter is quite useful to pinpoint the unconnected nodes. The process and the difference can be seen in the pictures below.

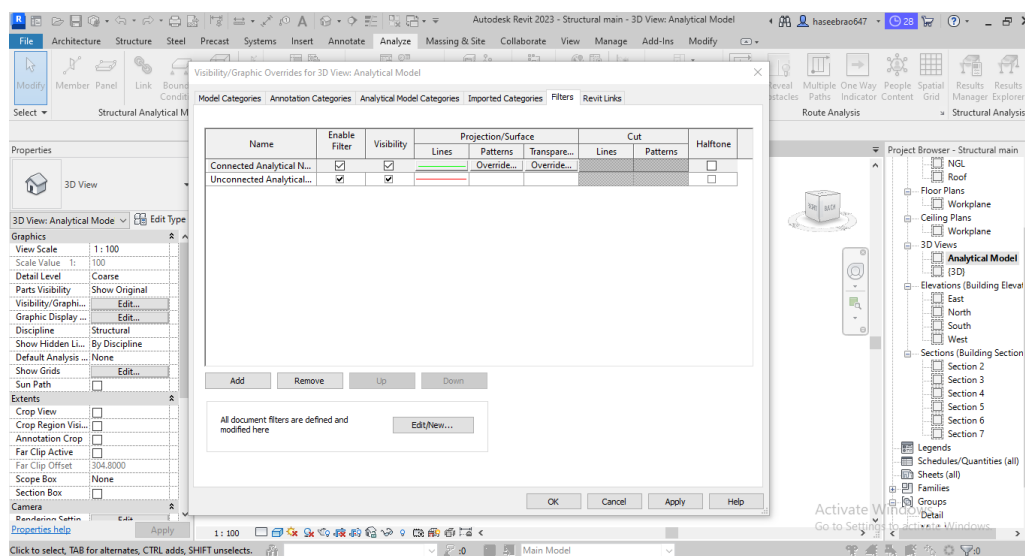


Figure 49: Filtering Analytical model for connected/unconnected Nodes

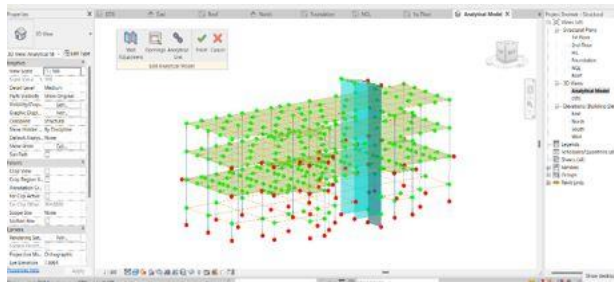


Figure 50: Unrefined/incorrect model based on Connections

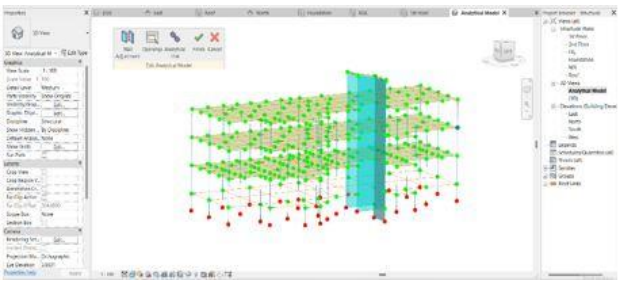


Figure 51: Corrected Analytical model

4. Architectural Design:

4.1. Loading Families

Command: Panel Insert, group Load from Library and command Load family

For brevity Conceptual model has been reused (duplicated) as basis for Architectural design. Afterwards for realistic touch, families of windows, doors, vents etc. were downloaded from “www.bimobject.com” website and introduced in the project through above command. Below there is an example of the dialog box in the case of loading the entrance door.

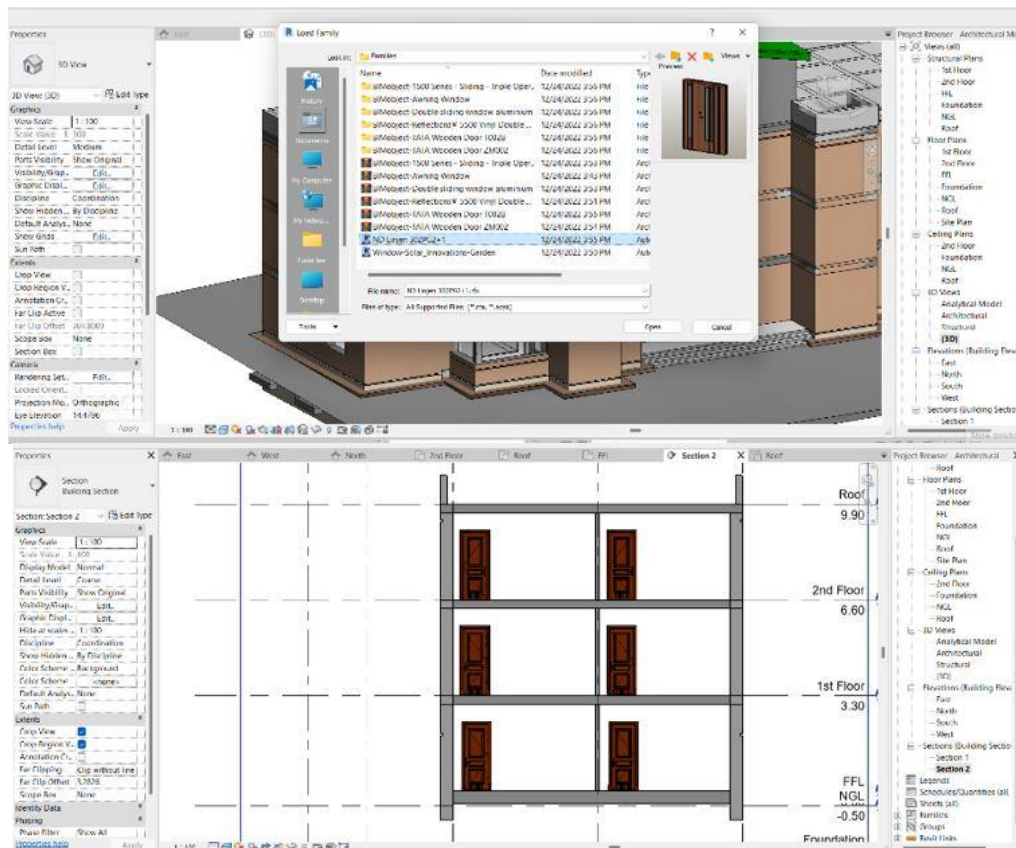


Figure 52: Loading Doors from Families

4.2. Material Management

Command: Properties window> edit type

As regards the management of materials, the example of the external Terrace wall is shown. By clicking on the material used for the finish layer it is possible, in the Appearance tab page, to change the material's color and the external finish. In this case Masonry stones were used to give it an aesthetic look and differentiable from the structure. Other useful properties for example costs etc. can also be associated to the material in this window which is quite useful for cost estimation (will be used later on in the project).

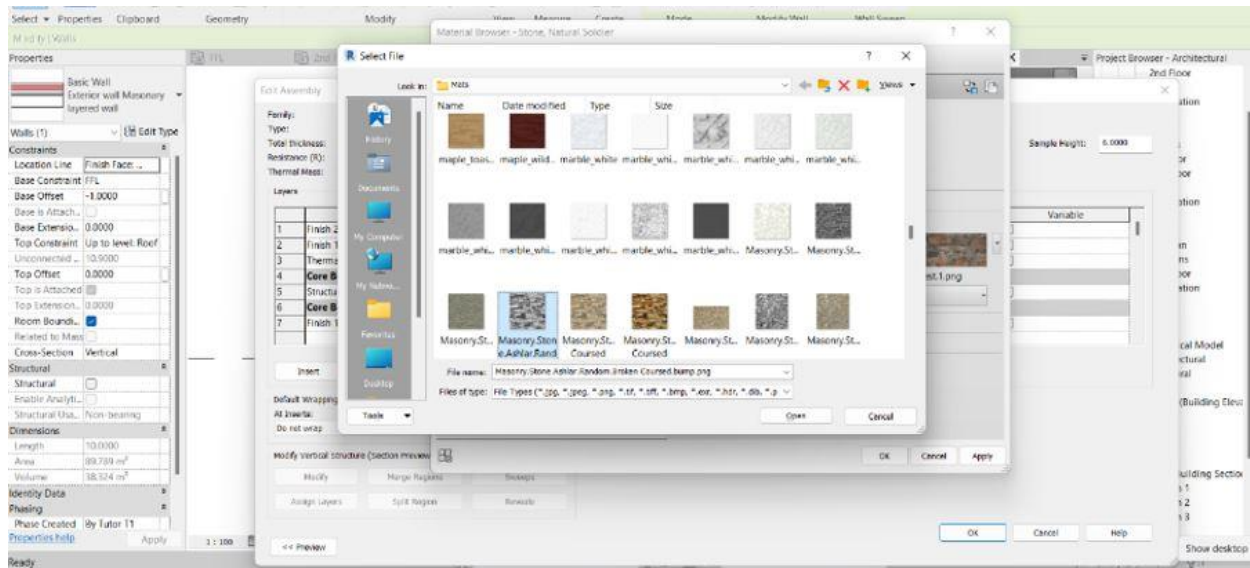


Figure 53: Properties Definition (To walls)

4.3. Object Refinements

Command: Properties window> edit type

This procedure was applied to change the single layering of the objects such as walls, roofs, and floors. The new architectural layers were created according to what was explained in the introduction chapter of the report as concerns the materials and the thickness, i.e. by selecting the object and clicking on the Edit Type command. So, for example, for the external wall, the current type of the wall was duplicated and renamed as "Exterior wall Brick Finish". Then the Edit Assembly window was opened, and new layers were added and modified in their function, material, thickness, and placement according to the core boundary. This same procedure was replicated also for the internal wall, the stairwell wall, the floor, and the roof refinements. In the next pictures the results are shown.

4.3.1. Floors

Starting from ground floor from bottom to up in the material layers on the site-hardcore, Rigid Insulation, Damp proofing were added as bottom layers. Then the core contained Concrete and a screeding layer. Afterwards Ceramic tiles were added to the Ground storey floor as finish material.

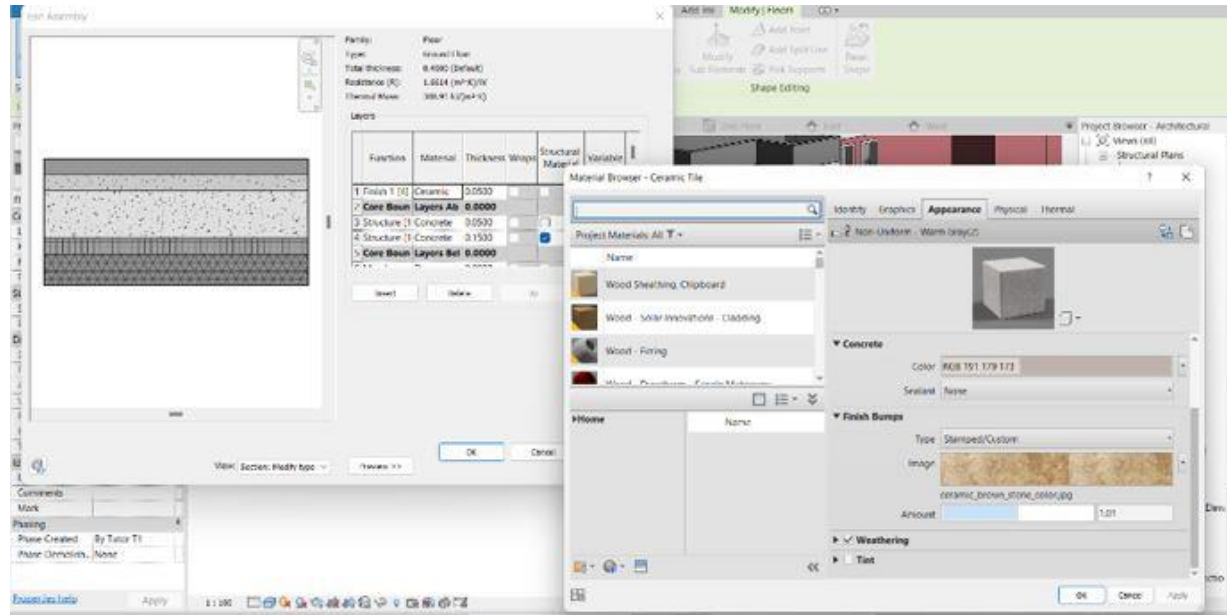


Figure 54: Ground Floor Refinement/Detailing

In the middle storeys, aside the 25mm slab and 5mm screeding, A membrane of polystyrene sheet and then ceramic tiles were added for finishing.

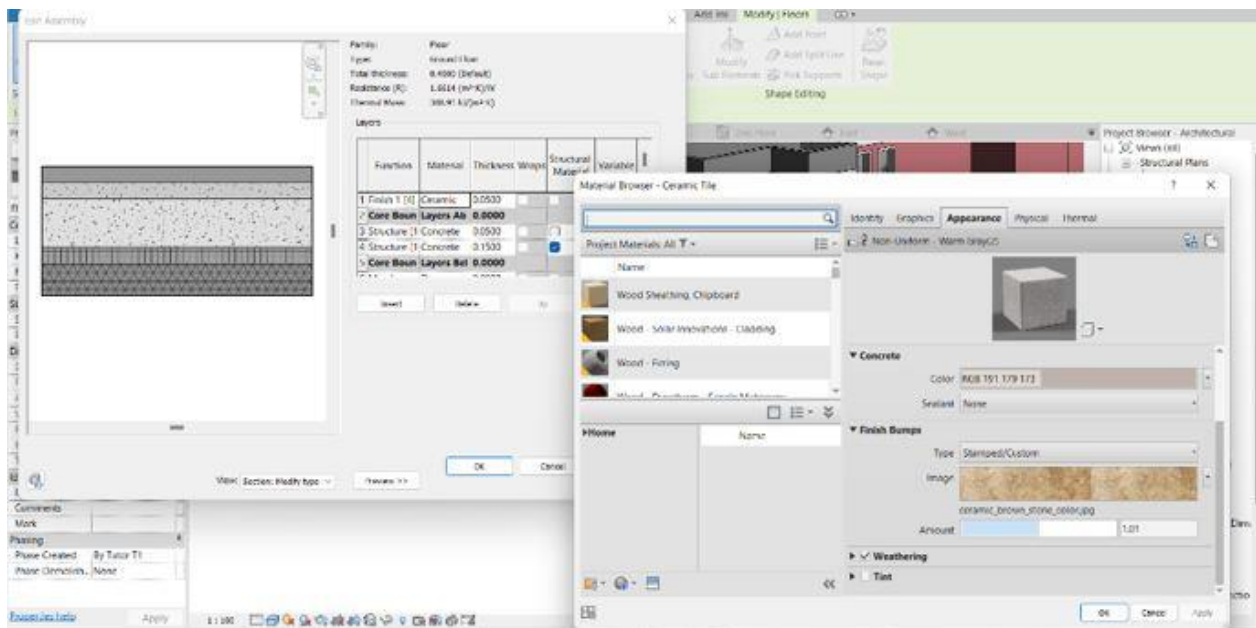


Figure 55: Mid Storey Floors Detail Refinement

For the top most slab (roof) same concrete slabs cores were finished off with Concrete walkway type finishing because of under usage of the topmost roof and cost control.

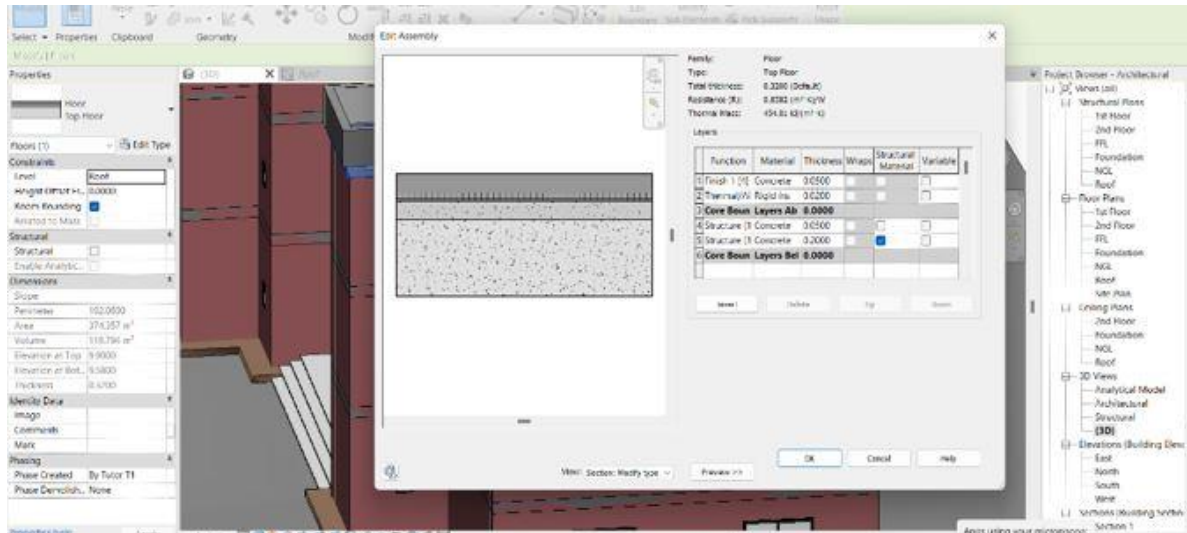


Figure 56: Top Roof Detail Refinement

4.3.2. Walls

All the walls contained masonry block elements made of concrete with their price assumed to be 50/sqm for cost scheduling purposes later.

Starting from the External walls concrete masonry blocks were used for core boundary with external finishing of common red brick types was assumed.

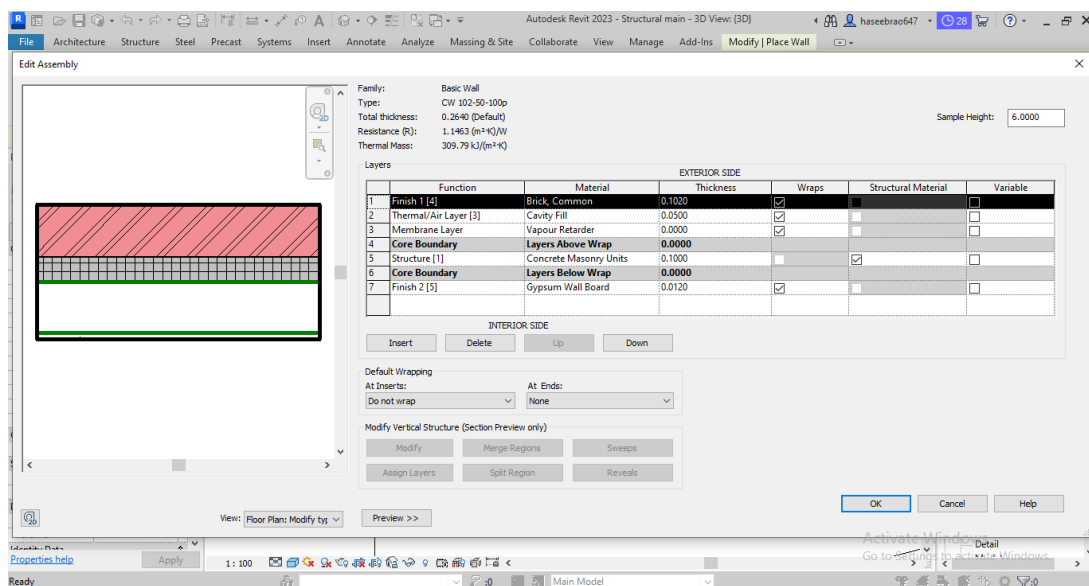


Figure 57: External Walls Detail Refinement

For the parapet walls (i.e. atop the roof at its boundaries) concrete blocks with adobe brick finishing was assumed.

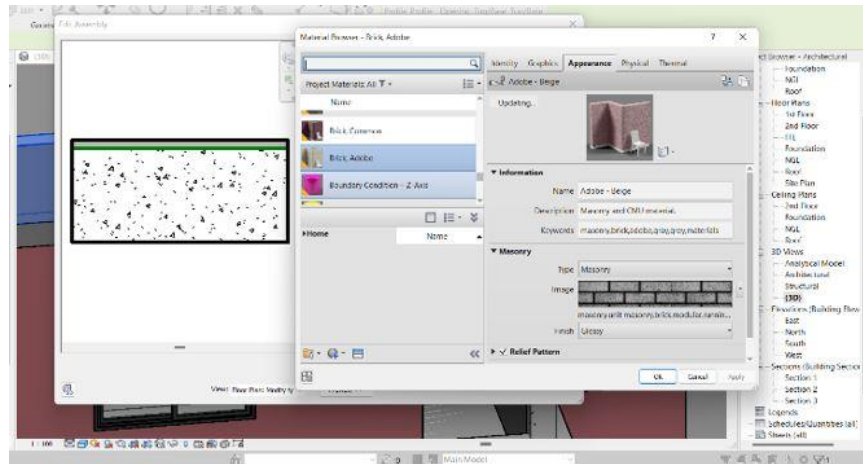


Figure 58: Parapet Wall Detail Refinement

For partition walls blocks enclosed in Gypsum wall boards were utilized.

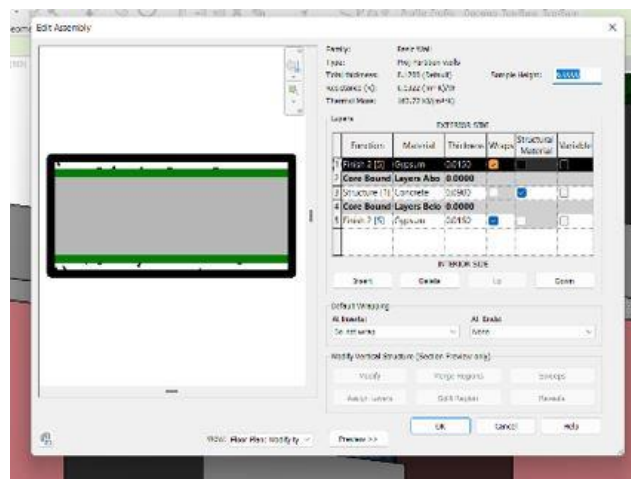


Figure 59: Partition wall detail Refinement

For Terrace wall blocks with stone finishing was used for aesthetics.

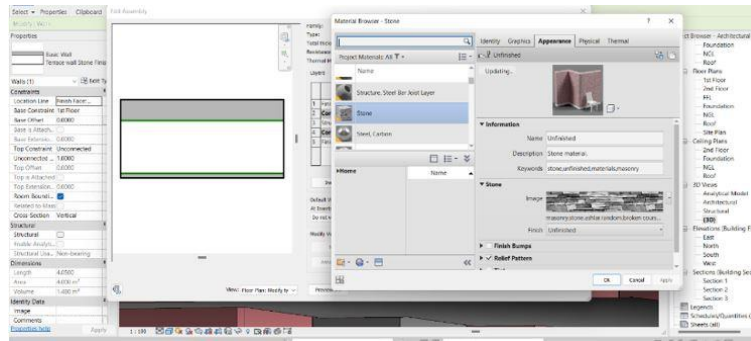


Figure 60: Terrace Wall Detail Refinement

4.4. Creating Families

Command: File > group New > command > Family template > English > Metric Window

With the Reference Plane command, additional planes are drawn in the Ref level view, which will define the references on which we will extrude.

Grey mosaic tiles were used as a material finish for the windows.

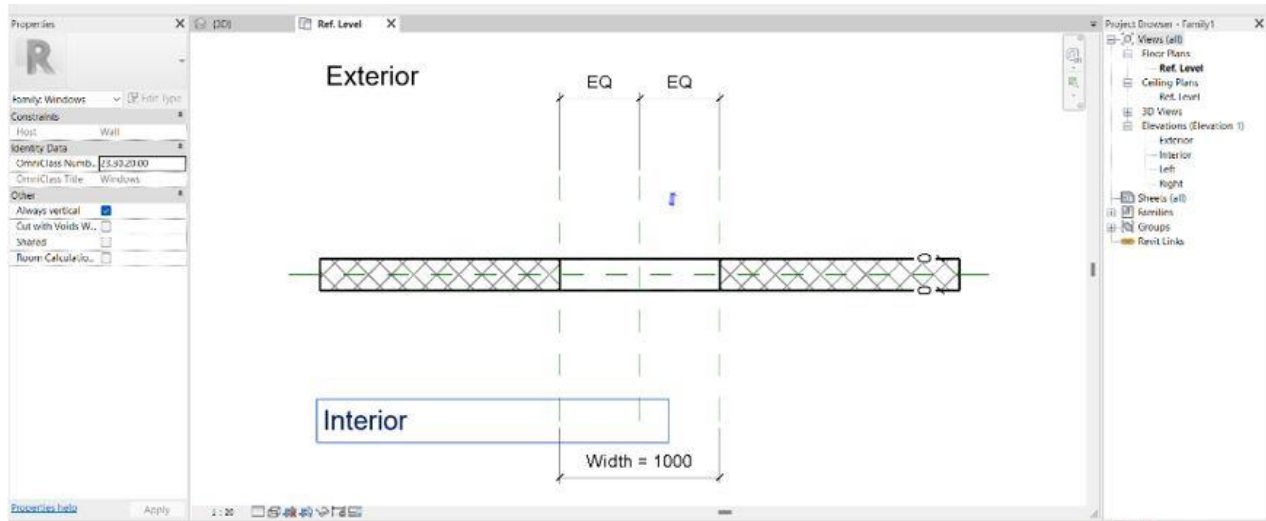


Figure 61: New Family Creation (Window)

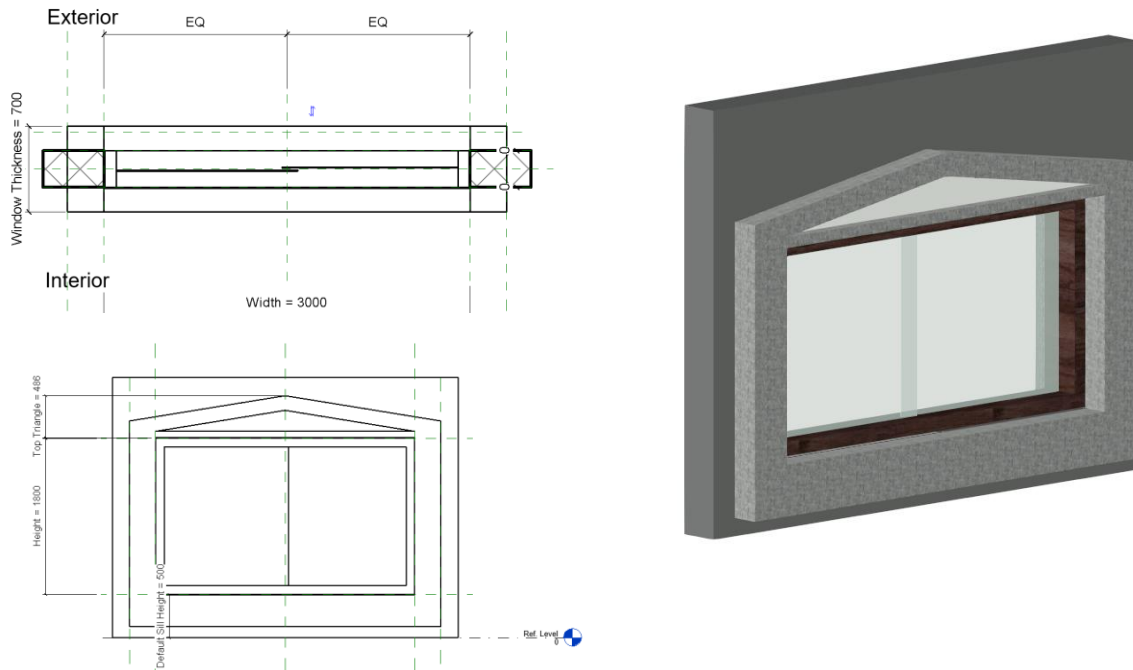


Figure 62: Creating Families (Extrusion Element Window)

4.5. Parameterizing Families

To make the family able to adapt to various conditions, the most significant dimensions are parameterized, so that they can be adjusted on will for those conditions. Here importantly the thickness of the window and the height of the triangle above the window were parameterized as they can sometime proof to be interfering with structure. A type parameter was assigned to the properties The required dimensions are shown in the following figure.

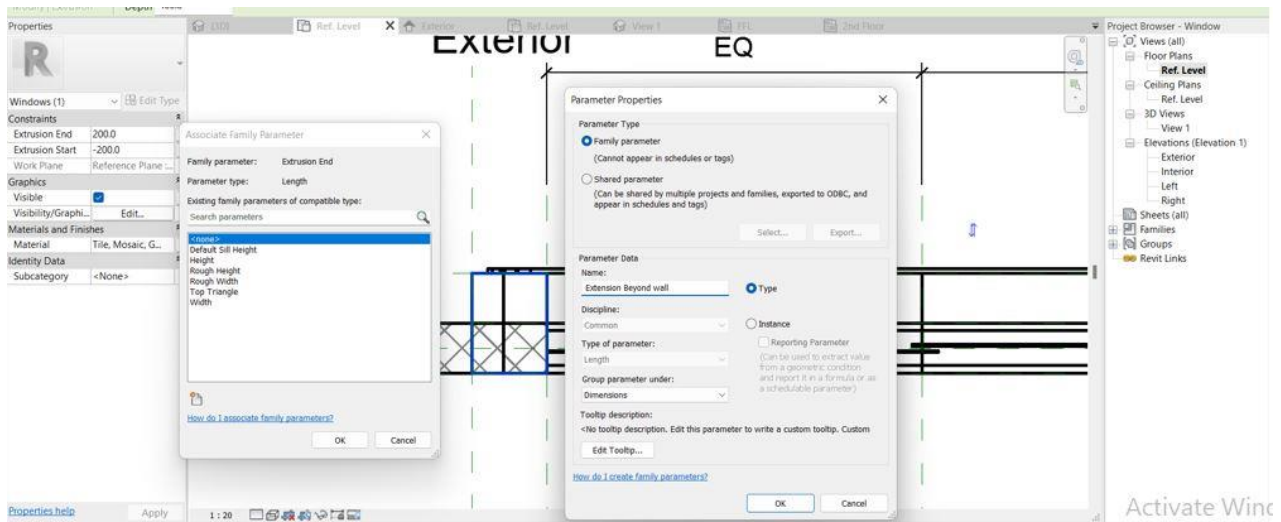


Figure 63: Parameterizing (Extension Beyond Wall)

The parameters can be added directly through the pen symbol on the dialogue box as shown in the figure.

All the parameters are defined therein.

The extension beyond the wall carried values in negative. (the sign is just indicative of extension going outside the wall's surface). The value were crossed check later in the project after loading.

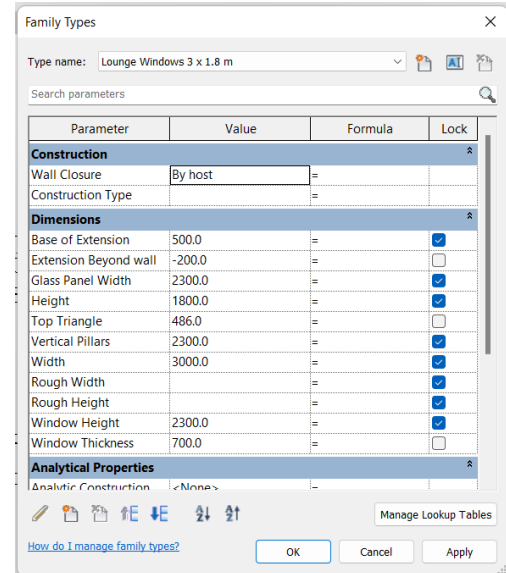
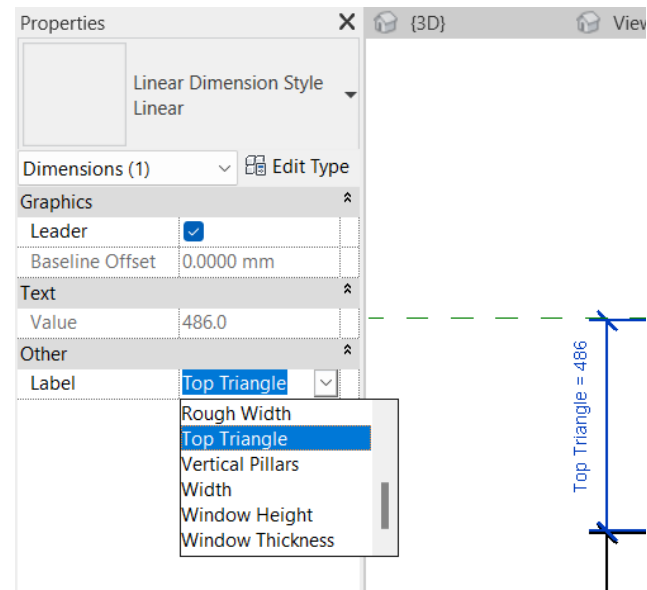


Figure 64: New Family Parameter Definitions

After the parameters are defined they can be attributed to the specific dimensions by selecting the dimension and choosing the specific label shown here.

Figure 65: Assigning Dimensions (parameterizing families)



The families were afterwards introduced into the project as shown below.

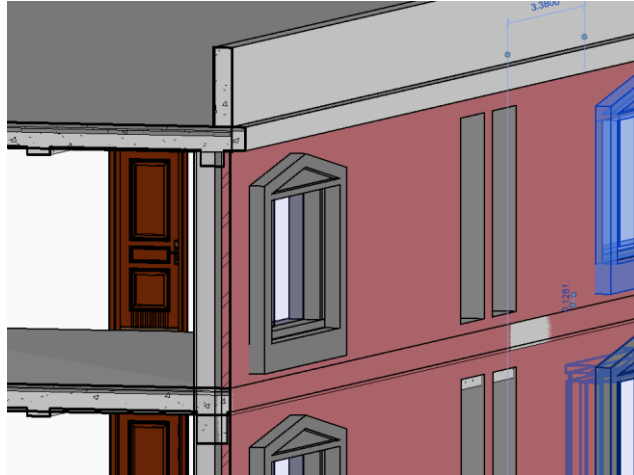


Figure 66: Introducing Families to Project

To verify the correctness of the parameterization, tests are carried out by changing the editable dimensions. To edit the type parameter, once selected the object in Edit Type it is possible modify the thickness of window etc.

4.6. Topographic Refinement

Command: Panel Massing & site

To make the project more aesthetically pleasing and complete topographic refinements were made on the site.

First a site plan was created on ground level as a basis for all the further detailing. This can be done by simply duplicating the view from the project browser windows. Afterwards the elements of topography were introduced in the project.

This includes first adding a building peripheral walkway of concrete with thickness of 0.3m atop a sand underlay. This marks the boundary between the ground and the building.



Figure 67: Creation of Sidewalk on Building Periphery

Afterwards Asphalt roads, were added by first using detail lines (present in Panel annotate Group Detail) to mark the extents on the topo surface already made before in conceptual design, These were in the form of curves and straight segments closed surfaces. Then the command split surface was used to split the topography into multiple segments to classify as roads, walkways, gardens etc.

Once the surfaces are split, they can be modified for their material usage by directly selecting them and using the edit type to select the material.

For aesthetics a new family was made namely curb stone but as a “railing” element. (File > Create families > English > Railing).

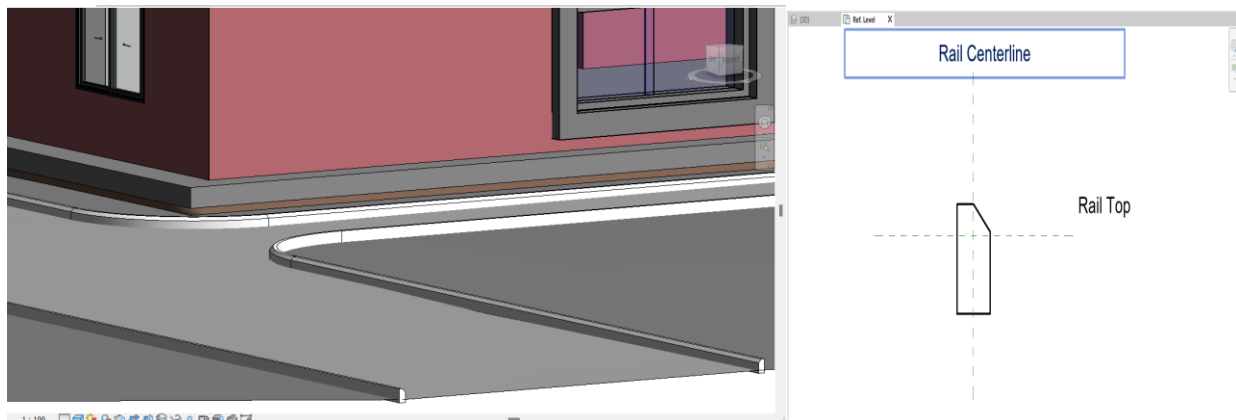
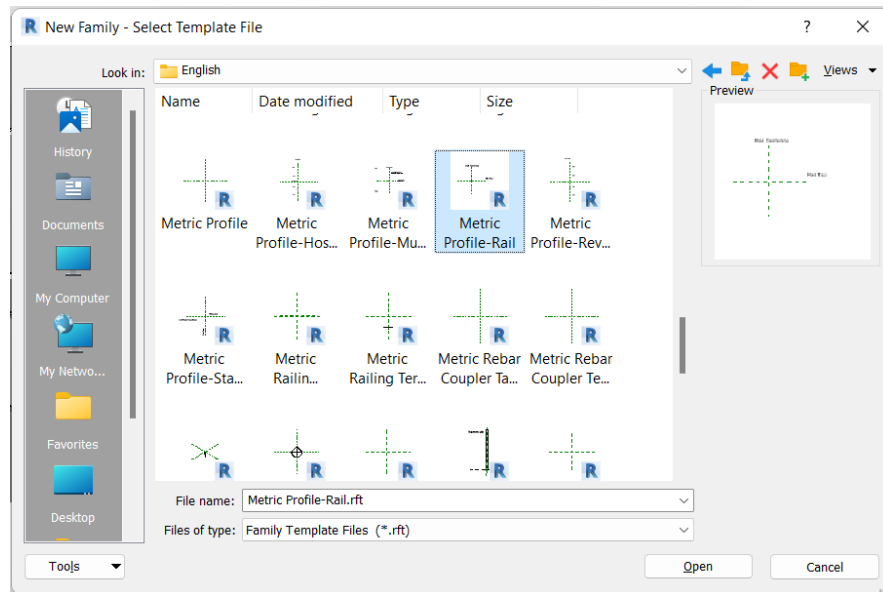


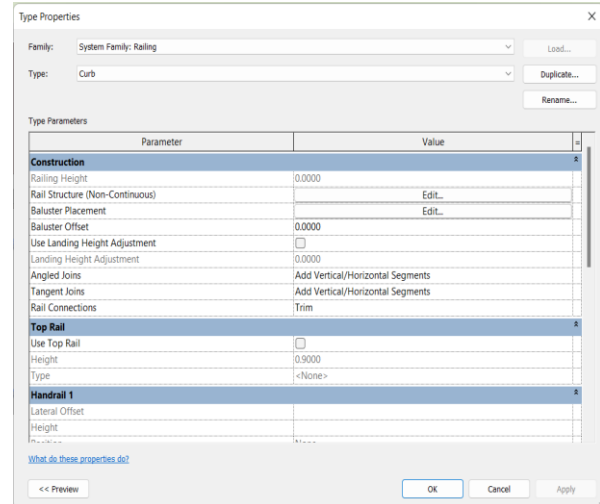
Figure 68: Creating New Curb Family

After this using the same detail lines previously made to split surface, Railing was introduced over these lines.

Changes to properties of railings were made after the curb railing was introduced.

Important of which were,

The Balusters family were all made to None and also the Top handrail was removed.



After some refinements, the following site plan & 3D view was obtained

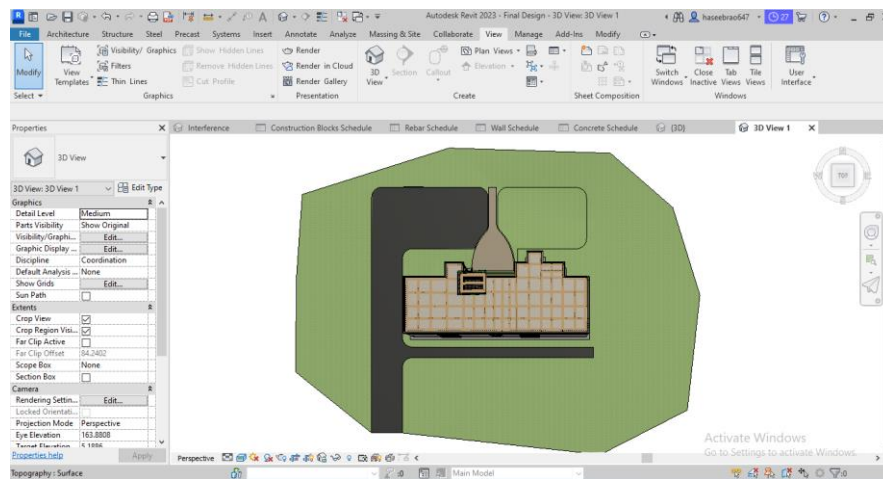


Figure 69: Site Plan View

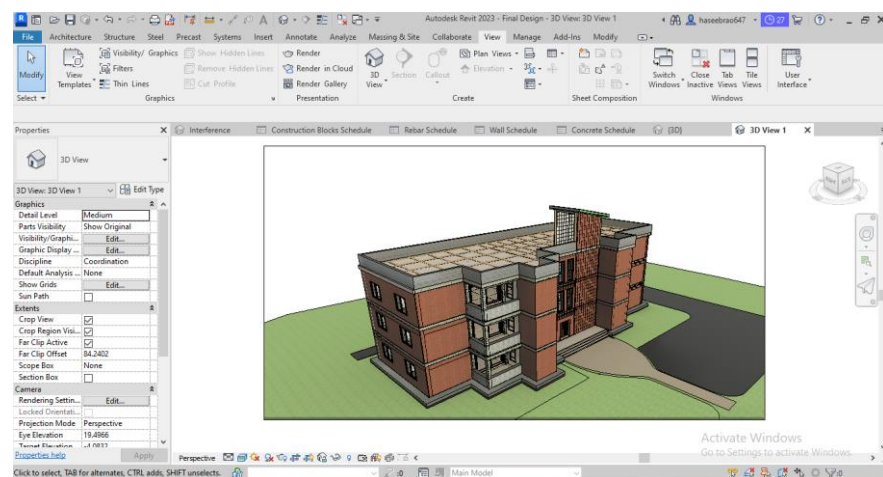
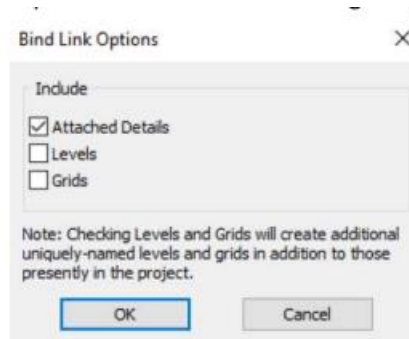


Figure 70: Final Site Layout

5. Final Design:

5.1. Federated Model Assembly

The last phase of the project consists of the union of both the architectural and structural. First a copy of the architectural model was made by saving it as “Final_Design.rvt”, After that, using the bind link command, once the structural model still linked in the project has been selected, join the “Structural_Design.rvt” and the “Architectural_Design.rvt”. During the procedure, in the “Bind Link Options” window, the check on the Attached details cell only is kept as shown in the following image.



After that the link associated with the structural model is deleted by entering Manage Link.

The Ungroup command is now used so that the individual elements initially present in the structural model can be selected.

5.2. Phases Management

Command: Panel Manage > group Phasing > command Phases

Phases define the essential milestones throughout the project timeline. In other words it's the sequence of construction. The procedure and the result of creating and managing the phases in the project are shown below.

Following the command, a total of 9 phases spanning the project were introduced which can be seen in the following figure.

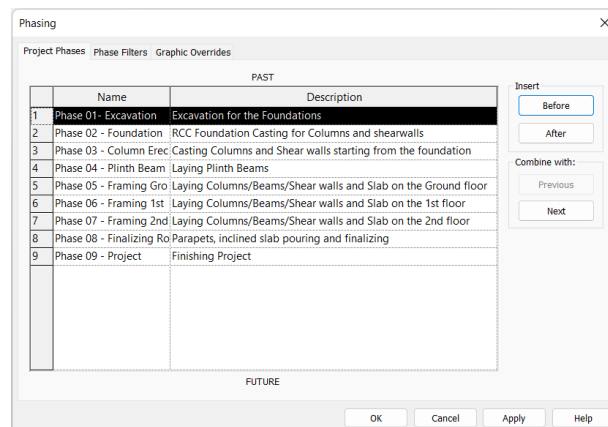


Figure 71: Introducing Phases

Certain overrides can be attributed to the phases based on their construction phase. As shown below.

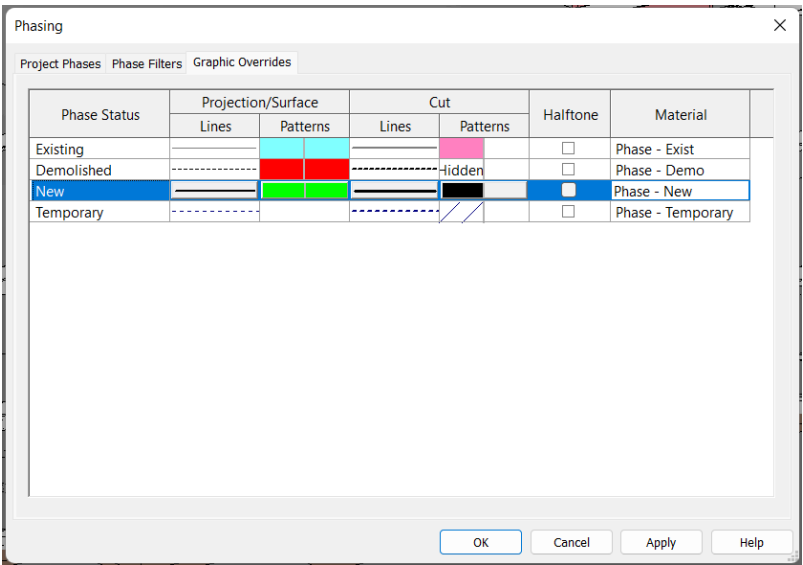


Figure 72: Graphic Overrides (Phases Management)

After that, the different elements of the project were given the corresponding Phase of construction and for each view that was created in the previous phase, the properties of visualization were set according to the corresponding phase of representation.

As an example, following is the comparison of Phase 6 (framing and Slab pouring 1st floor) and Phase 7 (framing and Slab pouring 2nd floor) is shown. The blue color indicates the existing structure (check overrides) while the grey shows the phase under consideration.

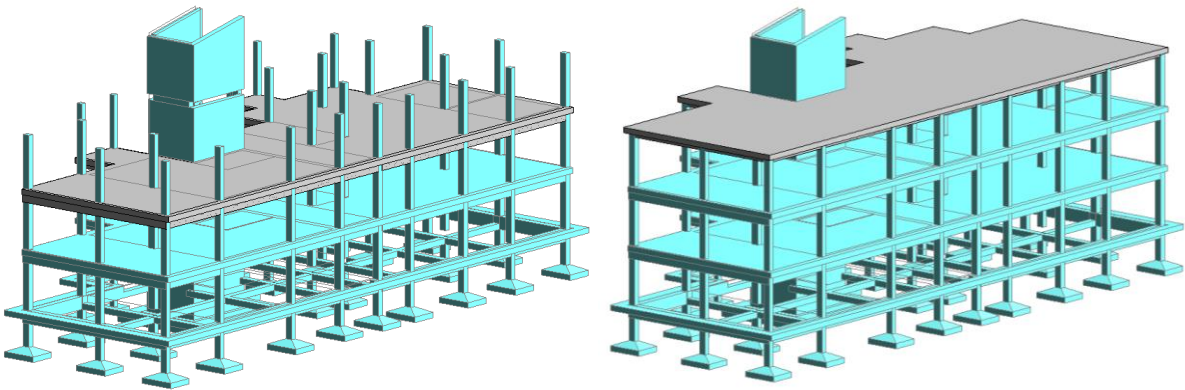


Figure 73: Phasing (Comparison of Phase - 6 and Phase 7)

5.3. Interference Check

Command: Panel Collaborate, group Coordinate, command Interference Check

After the combination of architectural and structural there were many elements where overlap occurs and can lead to material over estimation hence an interference check was performed between the two models by selecting all and checking. Run the interference.

Prior to final design a preliminary interference check was also made on structural project whereby check resulted in no interference.

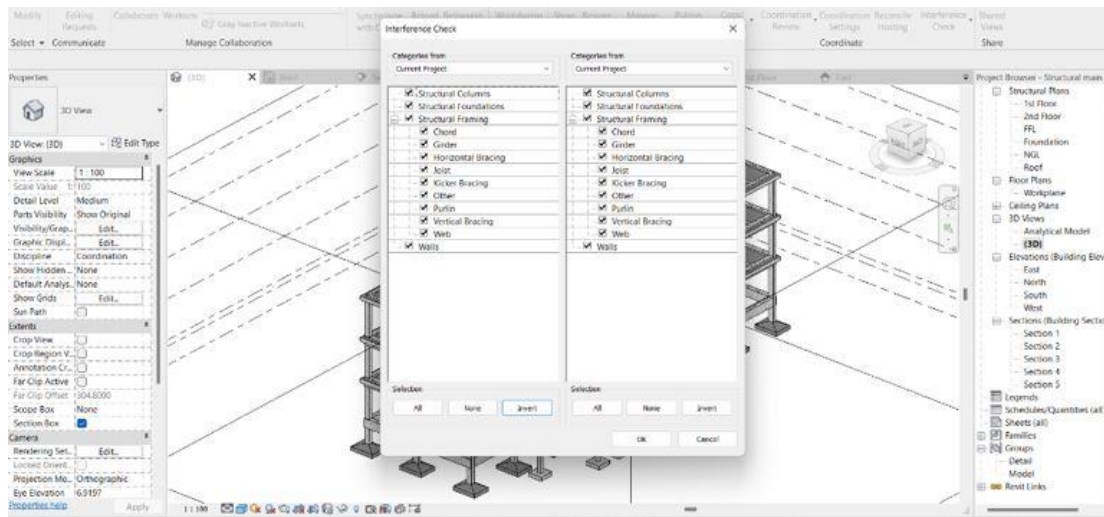


Figure 74: Interference Check (Structural to Structural)

But after the combination of structural and architectural models, Many interferences ranging from structural to architectural building components, structural to structural elements, Interferences of families with structural components etc. were obtained. Some of them can be seen here.

The first image shows the interference of doors with the slab. It meant the door was embedded in the slab and hence was to be a bit elevated.

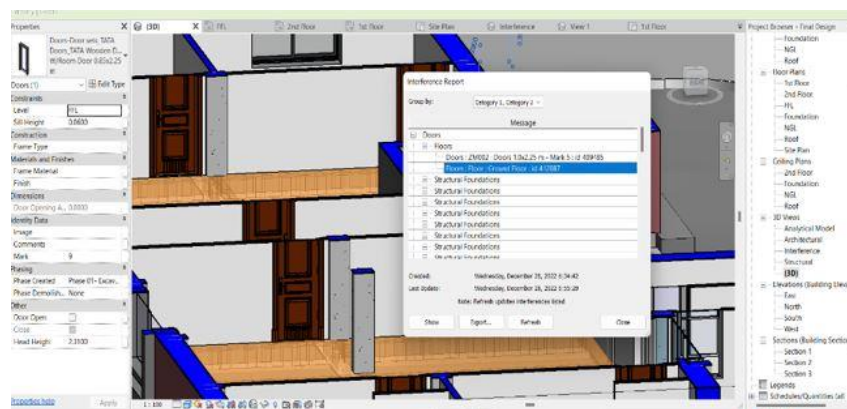


Figure 75: Interference Report (Final design - with issues-Pertaining to Doors with Structural Elements)

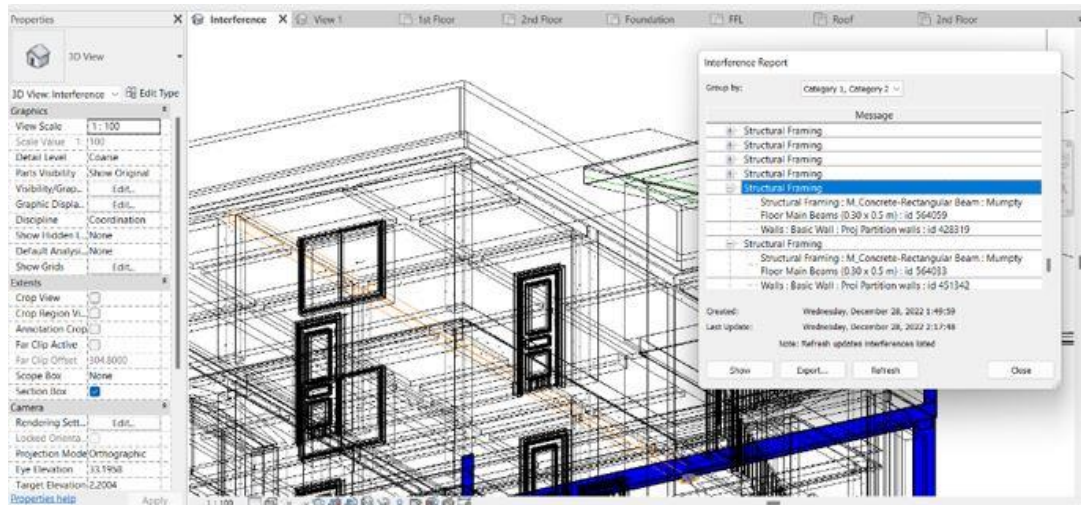


Figure 76: Interference Report (Final design - with issues-Pertaining to walls with Structural beams)



Figure 77: Interference Report (Final design - with issues-Pertaining to Floors with Structural Beams)

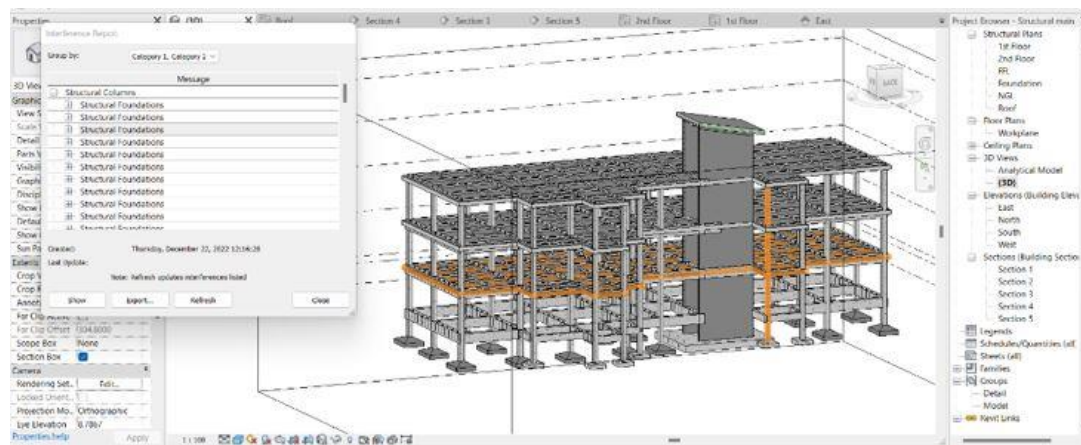


Figure 78: Interference Report (Final design - with issues-Pertaining to Slabs with Structural Columns)

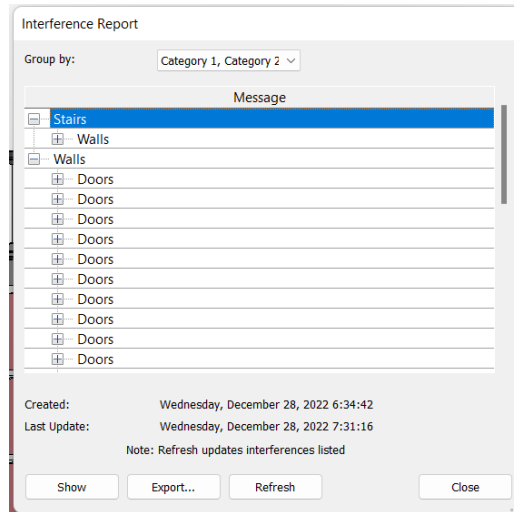


Figure 79: Final Interference Check

For what concerns downloaded families, such as windows, garage door or even stairway, their interferences have not been resolved because of their difficulties and time required. The remaining interferences are those shown in the following picture.

5.4. Elements Schedule

Command: Panel View > group Create > command Schedules > sub-command Schedule/Quantities

For the project Reinforcement details (Phase- 05) and surface of walls were scheduled.

For Reinforcement in Phase-05 that is the ground storey framing, the following fields were added,

- Bar Type
- Bar Diameter
- Shape
- Bar Length
- Count
- Family type

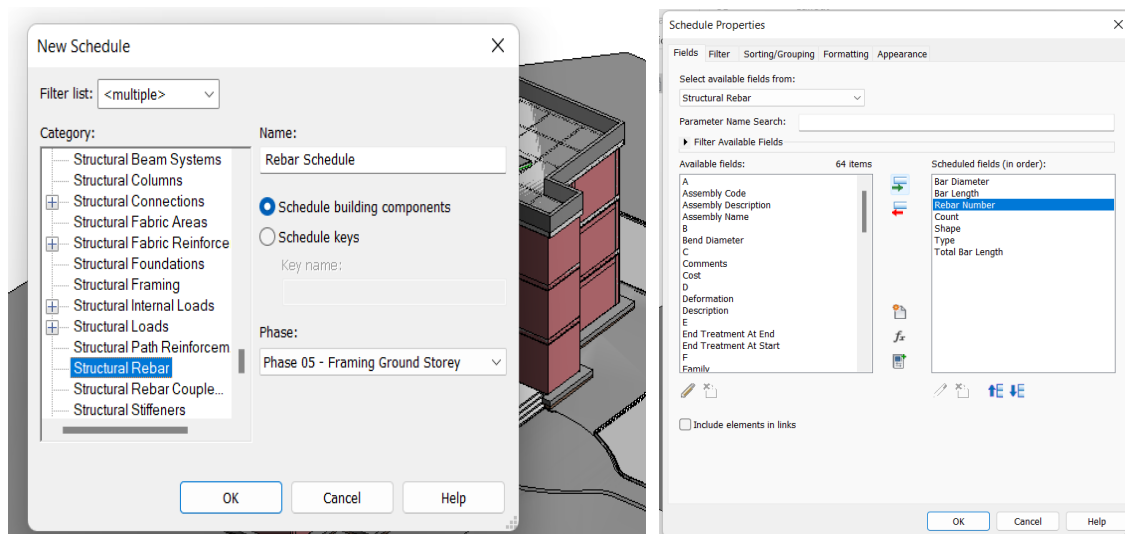


Figure 80: Reinforcement Schedule (Phase 5)

The schedule obtained is shown below. They were later sorted based on shapes.

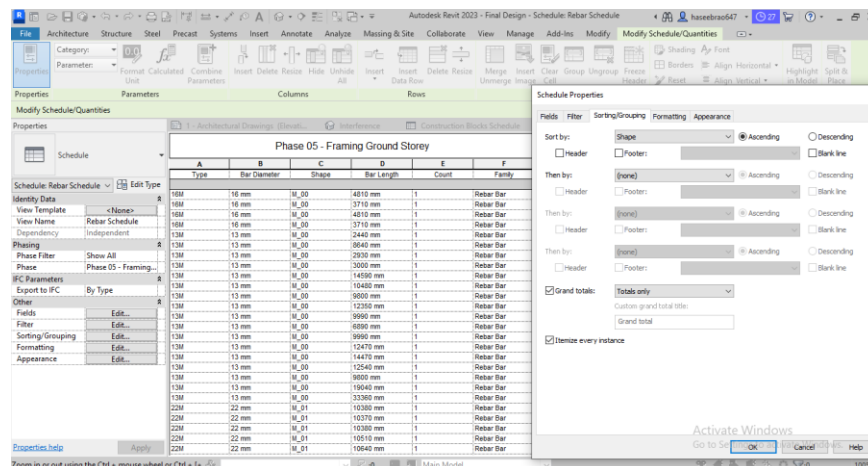


Figure 81: Creation of Rebar Schedules

- Family Type
- Structural material
- Width
- Unconnected Height
- Area

The screenshot displays the Autodesk Revit 2023 interface with the 'Schedule Properties' dialog box open for a 'Wall Schedule'. The dialog is configured with 'Family and Type' as the sorting method, and 'Footer' is checked with the title 'Title, count, and totals'. The 'Schedule' table lists various wall types and their quantities. The 'Properties' panel on the left shows the 'Schedule' properties, including 'View Template' set to '<None>' and 'View Name' set to 'Wall Schedule'.

Family and Type	Structural Material	C	D
Basic Wall: Basic Shear Wall (stairs)	Concrete, Cast-in-Place gray	0.34	13.40
Basic Wall: Basic Shear Wall (stairs): 2	Concrete, Cast-in-Place gray	0.34	14.78
Basic Wall: Basic Shear wall- (External-340mm)	Concrete, Cast-in-Place gray	0.34	11.90
Basic Wall: Basic Shear wall- (External-340mm): 1	Concrete, Cast-in-Place gray	0.34	11.90
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish	Concrete Masonry Units	0.47	10.20
Basic Wall: Exterior Wall Brick Finish: 14	Concrete Masonry Units	0.47	10.20
Basic Wall: Parapet wall	Concrete Masonry Units	0.25	34 m²
Basic Wall: Parapet wall	Concrete Masonry Units	0.25	10 m²
Basic Wall: Parapet wall	Concrete Masonry Units	0.25	4 m²

5.5. Material Takeoff schedule and Costs Computation

As compared to above process this schedule also includes the cost aspect of the material utilized in the project and hence gives an economic view of the project.

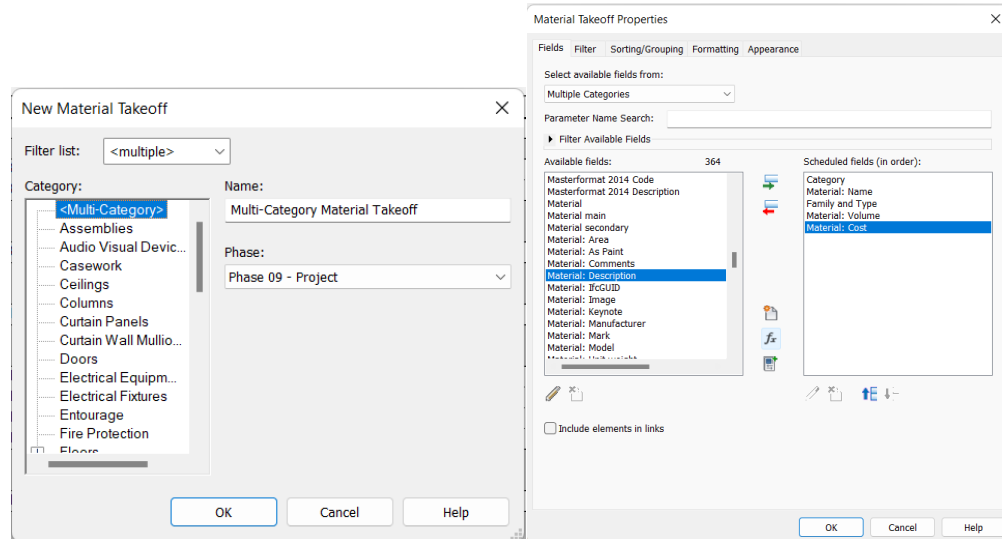
- Total Concreting (Cost = 130/m³)
- Total Concrete Masonry Units (Blocks) (Cost = 50 /m³) (assumed based on google search)

Before doing this however it was ascertained that all the structural members utilized “Concrete Cast in place, grey element.”

While “Concrete masonry Units” were utilized in all the wall elements (i.e. includes external walls, Partition walls, Terrace wall, Parapet)

5.5.1. Concrete

Firstly for the Concrete phase-09 was selected In multi categories added included “Material Name, category, family type, Material volume and material cost” .



This gives every type of material used in the project. They were then filtered based on their name to Concrete, Cast in place Grey as shown below.

The screenshot shows the Autodesk Revit 2023 interface with the 'Concrete Schedule' table open. The table has six columns: Material Name, Family and Type, Category, Material Volume, Material Cost, and Total Cost. The table lists various concrete elements, including walls, floors, and foundations, with their respective material volumes, costs, and total costs.

Material Name	Family and Type	Category	Material Volume	Material Cost	Total Cost
Concrete, Cast-in-Place gray	Basic Wall: Basic Shear Wall (stairs)	Walls	18.11 m³	130.00	2353.82
Concrete, Cast-in-Place gray	Basic Wall: Basic Shear Wall (stairs)	Walls	12.54 m³	130.00	1629.85
			30.64 m³		3983.67
Concrete, Cast-in-Place gray	Basic Wall: Basic Shear wall- (External-340mm)	Walls	10.56 m³	130.00	1372.56
			10.56 m³		1372.56
Concrete, Cast-in-Place gray	Floor: Ground Floor	Floors	49.14 m³	130.00	6388.52
			49.14 m³		6388.52
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	93.02 m³	130.00	12092.13
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	93.05 m³	130.00	12096.07
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	94.47 m³	130.00	12280.72
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	5.52 m³	130.00	717.27
			286.05 m³		37186.20
Concrete, Cast-in-Place gray	Foundation Slab: Mat foundation for Stairs (0.5 m)	Structural Foundations	19.25 m³	130.00	2502.50
			19.25 m³		2502.50
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.12 m³	130.00	15.63
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.11 m³	130.00	14.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.21 m³	130.00	26.80
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.39 m³	130.00	50.64
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.10 m³	130.00	12.96
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.11 m³	130.00	14.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extend	Structural Framing	0.11 m³	130.00	14.36

Figure 83: Concrete Schedule for Structural Elements

Once it was obtained a new field was created by name of total cost and checking the formula option. The formula input was a direct multiplication of Material Cost with Material Volume.

Refinements were made in how to view the schedule by sorting and grouping, adding a blank line and a footer with having the total Cost and Total volume row.

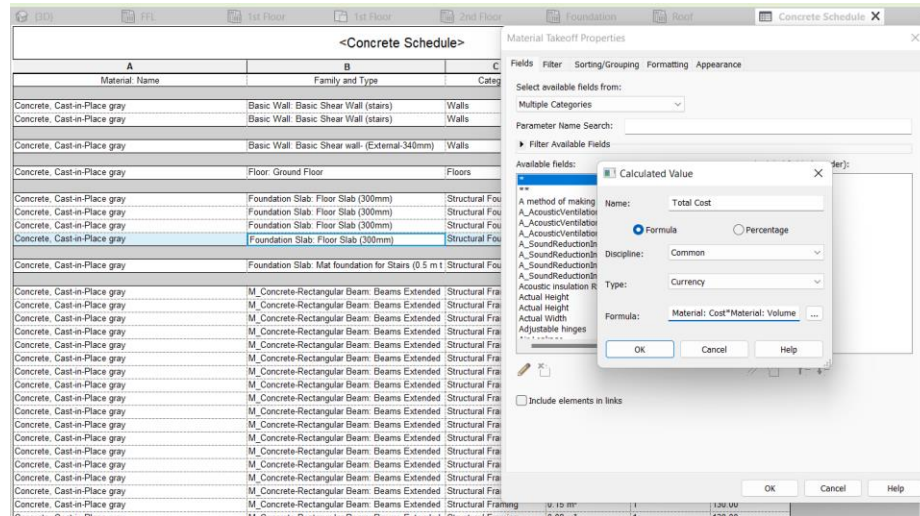


Figure 84: Adding Total Cost Field to Concrete Schedule

<Concrete Schedule>					
A	B	C	D	E	F
Material: Name	Family and Type	Category	Material: Volume	Material: Cost	Total Cost
Concrete, Cast-in-Place gray	Basic Wall: Basic Shear Wall (stairs)	Walls	18.11 m³	130.00	2353.82
Concrete, Cast-in-Place gray	Basic Wall: Basic Shear Wall (stairs)	Walls	12.54 m³	130.00	1629.85
			30.64 m³		3983.67
Concrete, Cast-in-Place gray	Basic Wall: Basic Shear wall- (External-340mm)	Walls	10.56 m³	130.00	1372.56
			10.56 m³		1372.56
Concrete, Cast-in-Place gray	Floor: Ground Floor	Floors	49.14 m³	130.00	6388.52
			49.14 m³		6388.52
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	93.02 m³	130.00	12092.13
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	93.05 m³	130.00	12096.07
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	94.47 m³	130.00	12280.72
Concrete, Cast-in-Place gray	Foundation Slab: Floor Slab (300mm)	Structural Foundations	5.52 m³	130.00	717.27
			286.05 m³		37186.20
Concrete, Cast-in-Place gray	Foundation Slab: Mat foundation for Stairs (0.5 m t)	Structural Foundations	19.25 m³	130.00	2502.50
			19.25 m³		2502.50
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.12 m³	130.00	15.63
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.11 m³	130.00	14.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.21 m³	130.00	26.80
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.39 m³	130.00	50.64
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.10 m³	130.00	12.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.11 m³	130.00	14.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.11 m³	130.00	14.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.12 m³	130.00	15.63
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.11 m³	130.00	14.36
Concrete, Cast-in-Place gray	M_Concrete-Rectangular Beam: Beams Extended	Structural Framing	0.21 m³	130.00	26.80

Figure 85: Take off schedule (concrete)

For the second material take-off Concrete masonry units (blocks) were shown. The fields included material description, Category, family and type, Material area and material cost.

This time the filter was made with description have concrete masonry units.

Quantity Schedule/Quantities

Properties

Schedule

Schedule: Constructio - Edit Type

Identify Data

View Template -<None>

View Name Construction Blo...

Dependency Independent

Phasing

Phase Filter: Show All

Phase Phase 09 - Project

Other

Fields Edit...

Filter

Sorting/Grouping Edit...

Formatting Edit...

Appearance Edit...

Properties help

Apply

File Edit Construction Blocks Schedule X

<Construction Blocks Schedule>

A	B	C	D	E	F
Material	Description	Category	Family and Type	Material Area	Material Cost
Concrete masonry	Walls	Basic Wall	Exterior 241 m²	53.00	12036
Concrete masonry	Walls	Basic Wall	Exterior 72 m²	50.00	3607
Concrete masonry	Walls	Basic Wall	Exterior 68 m²	50.00	3420
Concrete masonry	Walls	Basic Wall	Exterior 23 m²	50.00	1162
Concrete masonry	Walls	Basic Wall	Exterior 26 m²	50.00	1258
Concrete masonry	Walls	Basic Wall	Exterior 18 m²	50.00	902
Concrete masonry	Walls	Basic Wall	Exterior 33 m²	50.00	1616
Concrete masonry	Walls	Basic Wall	Exterior 72 m²	50.00	3607
Concrete masonry	Walls	Basic Wall	Exterior 39 m²	50.00	1968
Concrete masonry	Walls	Basic Wall	Exterior 26 m²	50.00	986
Concrete masonry	Walls	Basic Wall	Exterior 20 m²	50.00	996
Concrete masonry	Walls	Basic Wall	Exterior 16 m²	50.00	907
Concrete masonry	Walls	Basic Wall	Exterior 19 m²	50.00	909
Concrete masonry	Walls	Basic Wall	Exterior 22 m²	50.00	1124
602 m²					34080
Concrete masonry	Walls	Basic Wall	Parapet 34 m²	50.00	1692
Concrete masonry	Walls	Basic Wall	Parapet 10 m²	50.00	508
Concrete masonry	Walls	Basic Wall	Parapet 4 m²	50.00	221
Concrete masonry	Walls	Basic Wall	Parapet 3 m²	50.00	129
Concrete masonry	Walls	Basic Wall	Parapet 4 m²	50.00	200
Concrete masonry	Walls	Basic Wall	Parapet 2 m²	50.00	100
Concrete masonry	Walls	Basic Wall	Parapet 2 m²	50.00	112
Concrete masonry	Walls	Basic Wall	Parapet 2 m²	50.00	118
Concrete masonry	Walls	Basic Wall	Parapet 3 m²	50.00	141
Concrete masonry	Walls	Basic Wall	Parapet 2 m²	50.00	95
Concrete masonry	Walls	Basic Wall	Parapet 3 m²	50.00	129
Concrete masonry	Walls	Basic Wall	Parapet 3 m²	50.00	156
Concrete masonry	Walls	Basic Wall	Parapet 6 m²	50.00	394
Concrete masonry	Walls	Basic Wall	Parapet 2 m²	50.00	116
Concrete masonry	Walls	Basic Wall	Parapet 10 m²	50.00	496

Material Takeoff Properties

Fields Filter Sorting/Grouping Formatting Appearance

Sort by: Family and Type Ascending Descending

Header Footer Totals only Blank line

Then by: (none) Ascending Descending

Header Footer Blank line

Then by: (none) Ascending Descending

Header Footer Blank line

Then by: (none) Ascending Descending

Header Footer Blank line

Grand totals: Custom grand total Grand total

Benzify every instance

OK Cancel Help

Project Browser - Final Design

- NGL
- Roof
- Site Plan
- Ceiling Plans
- 2nd Floor
- Foundation
- NGL
- Roof
- 3D Views
- 3D View 1
- Analytical Model
- Architectural
- Interference
- Structural
- (3D)
- Elevations (Building Elev)
- East
- North
- South
- West
- Sections (Building Sectio
- Section 2
- Section 3
- Section 4
- Section 5
- Legends
- Schedules/Quantities (all
- Concrete Schedule
- Construction Blocks Sch
- Rebar Schedule
- Wall Schedule
- Sheets (all)
- families

Figure 86: Construction Blocks Schedule with cost (50/sq m)

Then a new field namely total cost was added which was the product of material area with material cost.

Refinements were made in how to view the schedule by sorting and grouping, adding a blank line and a footer with having the total Cost and Total volume row.

<Construction Blocks Schedule>					
A	B	C	D	E	F
Material: Description	Category	Family and Type	Material: Area	Material: Cost	Total Cost
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	241 m²	50.00	12036
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	72 m²	50.00	3607
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	68 m²	50.00	3420
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	23 m²	50.00	1162
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	25 m²	50.00	1258
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	18 m²	50.00	922
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	33 m²	50.00	1656
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	72 m²	50.00	3607
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	39 m²	50.00	1968
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	20 m²	50.00	988
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	20 m²	50.00	996
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	18 m²	50.00	907
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	19 m²	50.00	929
Concrete masonry units	Walls	Basic Wall: Exterior Wall Brick Finish	22 m²	50.00	1124
			692 m²		34580
Concrete masonry units	Walls	Basic Wall: Parapet wall	34 m²	50.00	1692
Concrete masonry units	Walls	Basic Wall: Parapet wall	10 m²	50.00	508
Concrete masonry units	Walls	Basic Wall: Parapet wall	4 m²	50.00	221
Concrete masonry units	Walls	Basic Wall: Parapet wall	3 m²	50.00	129
Concrete masonry units	Walls	Basic Wall: Parapet wall	4 m²	50.00	200
Concrete masonry units	Walls	Basic Wall: Parapet wall	2 m²	50.00	100
Concrete masonry units	Walls	Basic Wall: Parapet wall	2 m²	50.00	112
Concrete masonry units	Walls	Basic Wall: Parapet wall	2 m²	50.00	118
Concrete masonry units	Walls	Basic Wall: Parapet wall	3 m²	50.00	141
Concrete masonry units	Walls	Basic Wall: Parapet wall	2 m²	50.00	95
Concrete masonry units	Walls	Basic Wall: Parapet wall	3 m²	50.00	129
Concrete masonry units	Walls	Basic Wall: Parapet wall	3 m²	50.00	156
Concrete masonry units	Walls	Basic Wall: Parapet wall	8 m²	50.00	394
Concrete masonry units	Walls	Basic Wall: Parapet wall	2 m²	50.00	116

Figure 87: Take off schedule (Masonry Blocks)

5.6. Rooms Legend and Schedule

Command: Panel Architecture, group Room & Area.

To demarcate the living space into rooms, kitchens etc, room tags and room separator commands were used. The tag carried the name and the area of the room.

A legend was afterwards created next to the plan view by going to the panel Annotations, group Color fill and command Color Fill Legend. To conclude this part, a schedule is created with the following items:
- Name; - Perimeter; - Area.

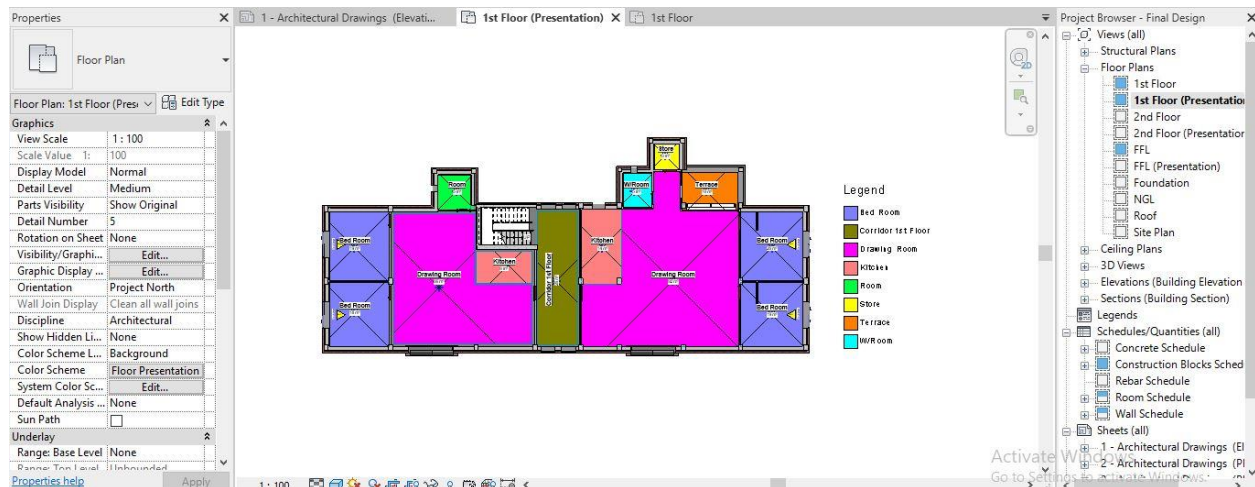


Figure 88: Room Legend (Ground Floor)

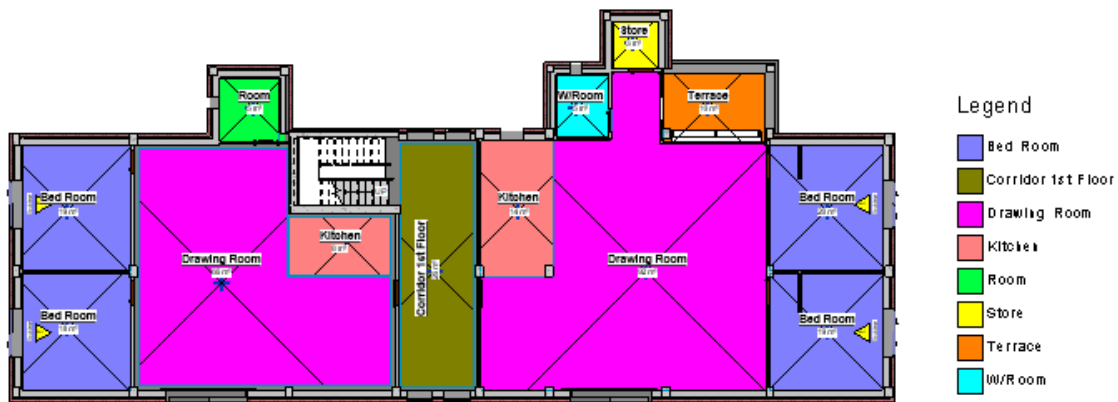


Figure 89: Room Legend (1st & 2nd Floor)

A total Room schedule for the whole project was also made as depicted under.

<Room Schedule>		
A	B	C
Name	Perimeter	Area
Bed Room	20.32	20 m²
Bed Room	19.58	18 m²
Bed Room	17.12	18 m²
Bed Room	17.68	19 m²
Bed Room	20.56	20 m²
Bed Room	20.48	19 m²
Bed Room	17.68	19 m²
Bed Room	17.12	18 m²
Bed Room	21.03	20 m²
Bed Room	20.48	19 m²
Bed Room	17.68	19 m²
Bed Room	17.12	18 m²

Figure 90: Room Schedule

5.7. Rendering

Final step involved rendering the 3D view. For the said purpose first a 3D layout was selected by placing the camera. Afterwards the quality was set to the best and the command was run.

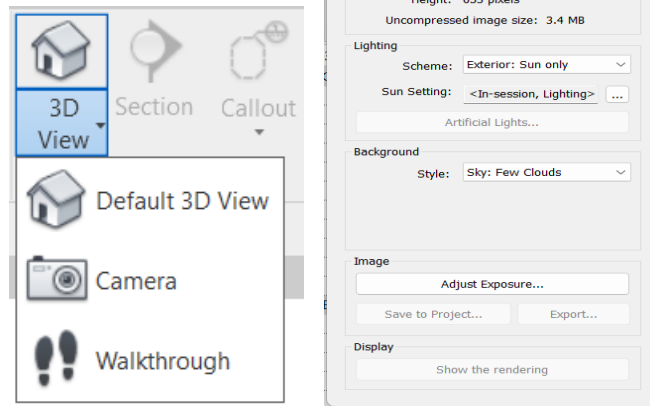


Figure 91: Rendering Properties

The final Rendered image is as follows.



Figure 92: Rendered Image

5.8. Sheets Generation

The last step of the project included the generation of sheets. A customized sheet was created building on the family A0_metric.rfa. Texts and images were inserted, and a new sheet was obtained. Afterwards the family was saved and loaded into the project.

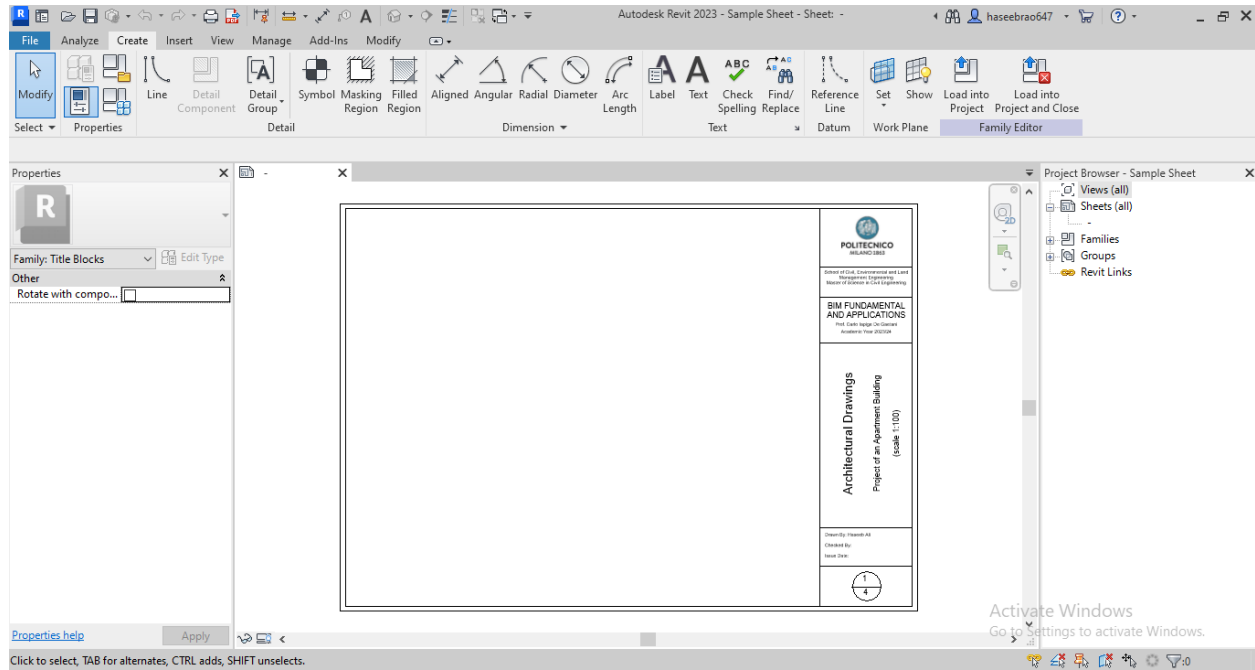


Figure 93: Sheets Generation

In the end four sheets were created encompassing the following elements.

SHEET NO.	SHEET NAME	SHEET ELEMENTS
1	Architectural Drawings (Elevations)	Includes 4 Elevations, Floor Presentation plan and Rendered image.
2	Architectural Drawings (Plans and x-sections)	2 Plans, 3 x-sections
3	Phasing	3D image, 8 Phases
4	Schedules	Concrete Masonry Units (Blocks) schedule, Room schedule, Wall Schedule



POLITECNICO
MILANO 1863

School of Civil, Environmental and Land
Management Engineering
Master of Science in Civil Engineering

BIM FUNDAMENTAL AND APPLICATIONS

Prof. Carlo Iapige De Gaetani
Academic Year 2023/24

Architectural Drawings

Project of an Apartment Building

(scale 1:100)

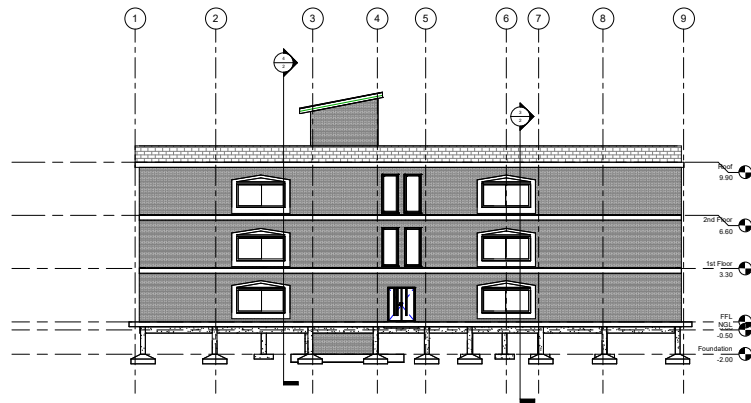
Drawn By: Haseeb Ali

Checked By:

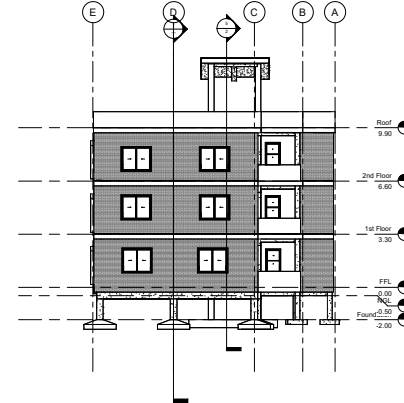
Issue Date:

2

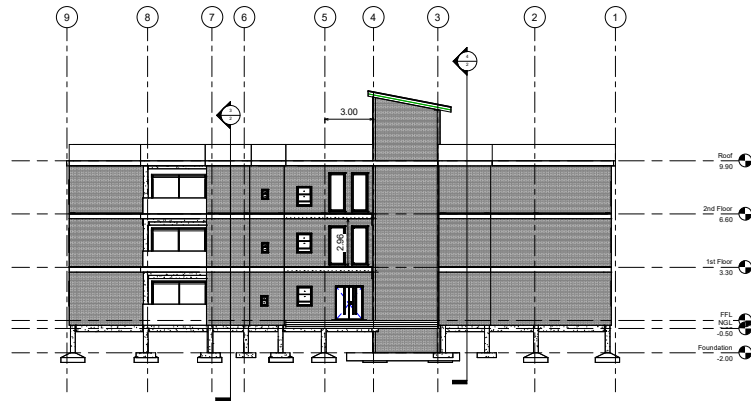
4



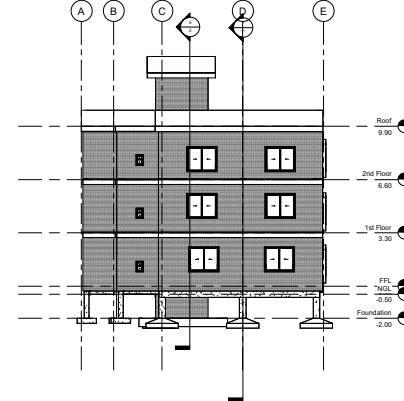
① South
1:100



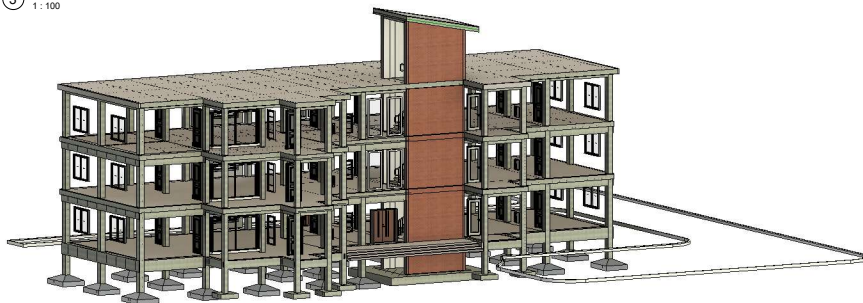
② East
1:100



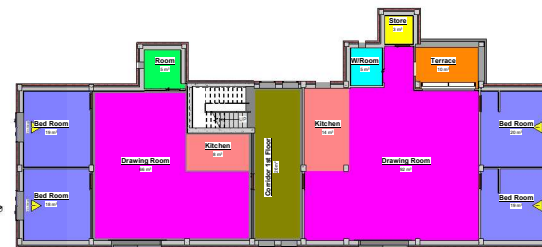
③ North
1:100



④ West
1:100



⑥ Architectural



⑤ 1st Floor (Presentation)
1:100



POLITECNICO
MILANO 1863

School of Civil, Environmental and Land
Management Engineering
Master of Science in Civil Engineering

BIM FUNDAMENTAL AND APPLICATIONS

Prof. Carlo Iapige De Gaetani
Academic Year 2023/24

Architectural Drawings

Project of an Apartment Building

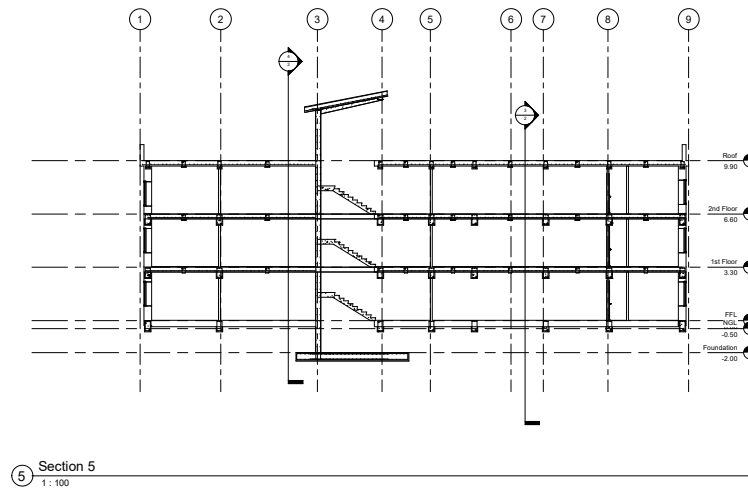
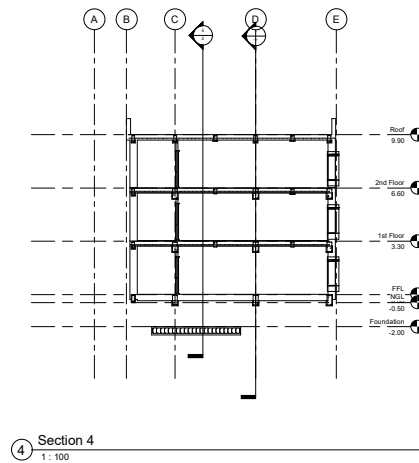
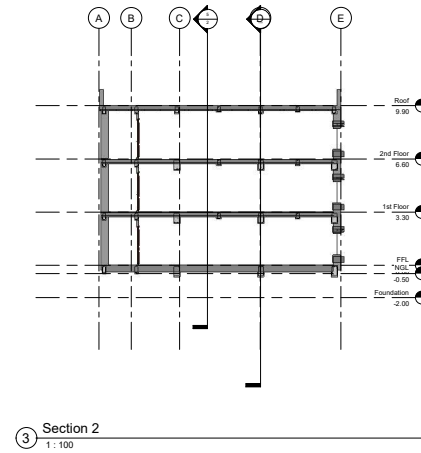
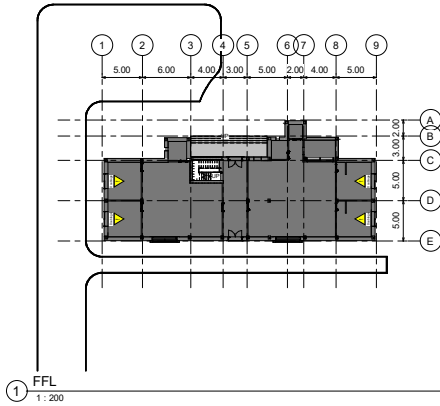
(scale 1:100)

Drawn By: Haseeb Ali

Checked By:

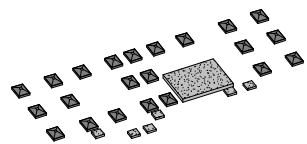
Issue Date:

2
4

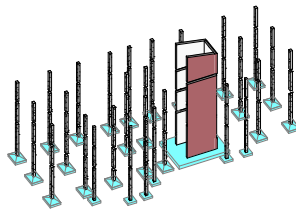




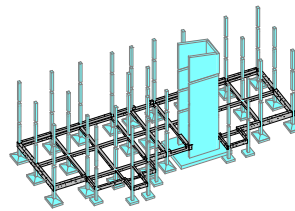
① (3D)



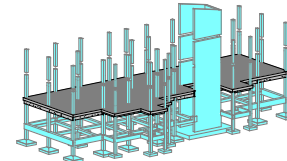
② Phase-2



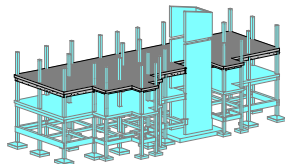
③ Phase-3



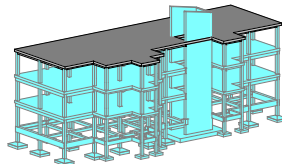
④ Phase-4



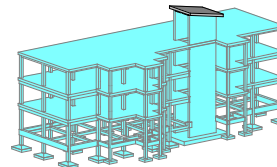
⑤ Phase-5



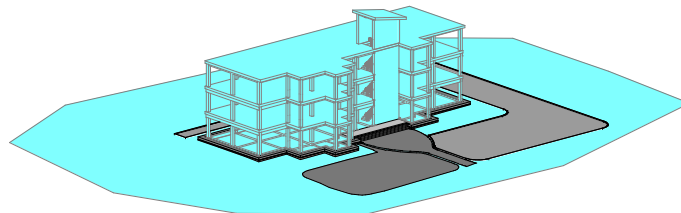
⑥ Phase-6



⑦ Phase-7



⑧ Phase-8



⑨ Phase-9



POLITECNICO
MILANO 1863

School of Civil, Environmental and Land
Management Engineering
Master of Science in Civil Engineering

BIM FUNDAMENTAL AND APPLICATIONS

Prof. Carlo Iapige De Gaetani
Academic Year 2023/24

Architectural Drawings

Project of an Apartment Building

(scale 1:100)

Drawn By: Haseeb Ali

Checked By:

Issue Date:

2

4

Construction Blocks Schedule					
Material Description	Category	Family and Type	Material Area	Material Cost	Total Cost
Concrete masonry units	Walls	Basic Wall Proj Partition walls	20 m²	50.00	1003
Concrete masonry units	Walls	Basic Wall Proj Partition walls	11 m²	50.00	568
Concrete masonry units	Walls	Basic Wall Proj Partition walls	12 m²	50.00	579
Concrete masonry units	Walls	Basic Wall Proj Partition walls	6 m²	50.00	283
Concrete masonry units	Walls	Basic Wall Proj Partition walls	4 m²	50.00	187
Concrete masonry units	Walls	Basic Wall Proj Partition walls	4 m²	50.00	221
Concrete masonry units	Walls	Basic Wall Proj Partition walls	4 m²	50.00	188
Concrete masonry units	Walls	Basic Wall Proj Partition walls	4 m²	50.00	224
Concrete masonry units	Walls	Basic Wall Proj Partition walls	16 m²	50.00	805
			405 m²		20241
Concrete masonry units	Walls	Basic Wall Terrace wall Stone Finish	3 m²	50.00	153
Concrete masonry units	Walls	Basic Wall Terrace wall Stone Finish	5 m²	50.00	264
Concrete masonry units	Walls	Basic Wall Terrace wall Stone Finish	3 m²	50.00	143
Concrete masonry units	Walls	Basic Wall Terrace wall Stone Finish	4 m²	50.00	212
Concrete masonry units	Walls	Basic Wall Terrace wall Stone Finish	3 m²	50.00	129
Concrete masonry units	Walls	Basic Wall Terrace wall Stone Finish	4 m²	50.00	203
			22 m²		1105

[illegible]

POLITECNICO
MILANO 1863

School of Civil, Environmental and Land
Management Engineering
Master of Science in Civil Engineering

BIM FUNDAMENTAL AND APPLICATIONS

Prof. Carlo Iapige De Gaetani
Academic Year 2023/24

Architectural Drawings

Project of an Apartment Building

(scale 1:100)

Drawn By: Haseeb Ali

Checked By:

Issue Date: